

UNIVERSIDAD EAFIT
School of Applied Sciences and Engineering
Department of Mechanical Engineering
IM0240 – Machine Design I

Drawing package

Tripteron CNC router machine

Delivery 2 – English cover and drawing index

Group 2

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Course lecturer

Juan Andrés Gallego Sánchez

Medellín, Colombia – 3 March 2024

1. About this package

This package contains the complete set of manufacturing and assembly drawings of the Tripteron CNC router. The drawing sheets that follow this index are the CAD sheets as issued from SolidWorks; they are reproduced unchanged so that the dimensions, tolerances and revision state shown on them remain the ones the parts were made to.

General notes applying to every sheet:

- **All dimensions are in millimetres (mm).** Angles are in degrees.
- Projection: first-angle (ISO-A).
- Sheet format: A3 unless stated otherwise on the sheet.
- Unless otherwise indicated: linear tolerance ± 0.5 mm, angular tolerance $\pm 0.5^\circ$, break all sharp edges.
- Do not scale the drawing; work from the stated dimensions.
- Materials are stated in the title block of each sheet; the structural members are steel tube and plate, the links are $25 \times 25 \times 2$ mm square steel tube.

2. Drawing index

Sheet	Title	Drawing no.	Scale	Drawn by	Status
Group 1 – Chassis and structure					
1	Structural table – general drawing	ME-001	1:10	D. Zuluaga Henao	redrawn
2	Sheet metal plate – table top	ME-002	1:5	D. Zuluaga Henao	redrawn
3	Top support bracket	ME-003	1:2	D. Zuluaga Henao	redrawn
4	Motor support	ME-004	1:2	D. Zuluaga Henao	redrawn
5	Pulley plate – quarter sheet	ME-005	1:2	D. Zuluaga Henao	redrawn
6	Motor side support	ME-006	1:1	D. Zuluaga Henao	redrawn
7	Top support bracket 2	ME-007	1:1	D. Zuluaga Henao	redrawn
8	Support bracket 2	ME-008	1:2	D. Zuluaga Henao	redrawn
Group 2 – Electrical system					
9	Electrical box cover plate	ME-009	1:2	D. Zuluaga Henao	redrawn
10	Wiring diagram – motor control system	EE-001	N/A	D. Zuluaga Henao	redrawn
11	Motor control schematic	EE-001	N/A	D. Zuluaga Henao	redrawn
12	Electrical bill of materials	EE-002	N/A	D. Zuluaga Henao	redrawn
13	Stepper motor driver wiring diagram	EE-003	N/A	D. Zuluaga Henao	redrawn
Group 3 – Guide-supported axis system					
14	CNC router – guide assembly	ME-001	1:3	J. M. Rodríguez	as issued
15	Guide 1 – general views	ME-002	1:4	J. M. Rodríguez	as issued
16	Guide 1 – exploded view and component list	ME-003	1:3	J. M. Rodríguez	as issued
17	Guide rail assembly	ME-003	1:2	G. Riaño	as issued
18	Guide 2 – general views	ME-004	1:3	G. Riaño	as issued
19	Guide 3 – general views	ME-005	1:3	G. Riaño	as issued
20	Guide carriage – isometric view	–	2:1	G. Riaño	as issued

Sheet	Title	Drawing no.	Scale	Drawn by	Status
21	Guide carriage – views	ME-006	1:2	G. Riaño	as issued
22	Bearing support – views	ME-007	1:1	J. M. Rodríguez	as issued
Group 4 – Arms (R-pair system)					
23	Arm assembly – component list	–	1:2	R. Bueno	as issued
24	Elbow and bearing assembly – section A–A	–	2:1	D. Zuluaga Henao	as issued
25	Bearing housing tube Ø30, gauge 16	–	2:1	R. Bueno	as issued
26	Tube Ø30, gauge 16	–	2:1	R. Bueno	as issued
27	Arm – fish-mouth cut, 1 in, 90°	–	1:2	R. Bueno	as issued
28	Milling tool support	–	2:1	D. Amell Osorio	as issued
29	Elbow assembly 1 – section A–A and component list	–	1:10	D. Zuluaga Henao	as issued
Group 5 – Assembled machine					
30	Complete machine – general views	–	1:10	G. Riaño Barrera	as issued
31	Complete machine – exploded view and part list	–	1:10	D. Amell Osorio	as issued
32	Washer (spacer)	–	10:1	D. Zuluaga Henao	as issued
33	Internal bearing spacer	–	5:1	D. Zuluaga Henao	as issued
34	Bearing housing tube	–	2:1	D. Zuluaga Henao	as issued
35	Solid tube with bore	–	2:1	D. Zuluaga Henao	as issued

Status says how each sheet reaches this package. *Redrawn* means the sheet was drawn again with an English title block and English notes, keeping the geometry and the dimensions of the original; *as issued* means the sheet is reproduced exactly as it was submitted on 3 March 2024, with its Spanish title block. Thirteen of the thirty-five sheets were redrawn; the rest are untouched.

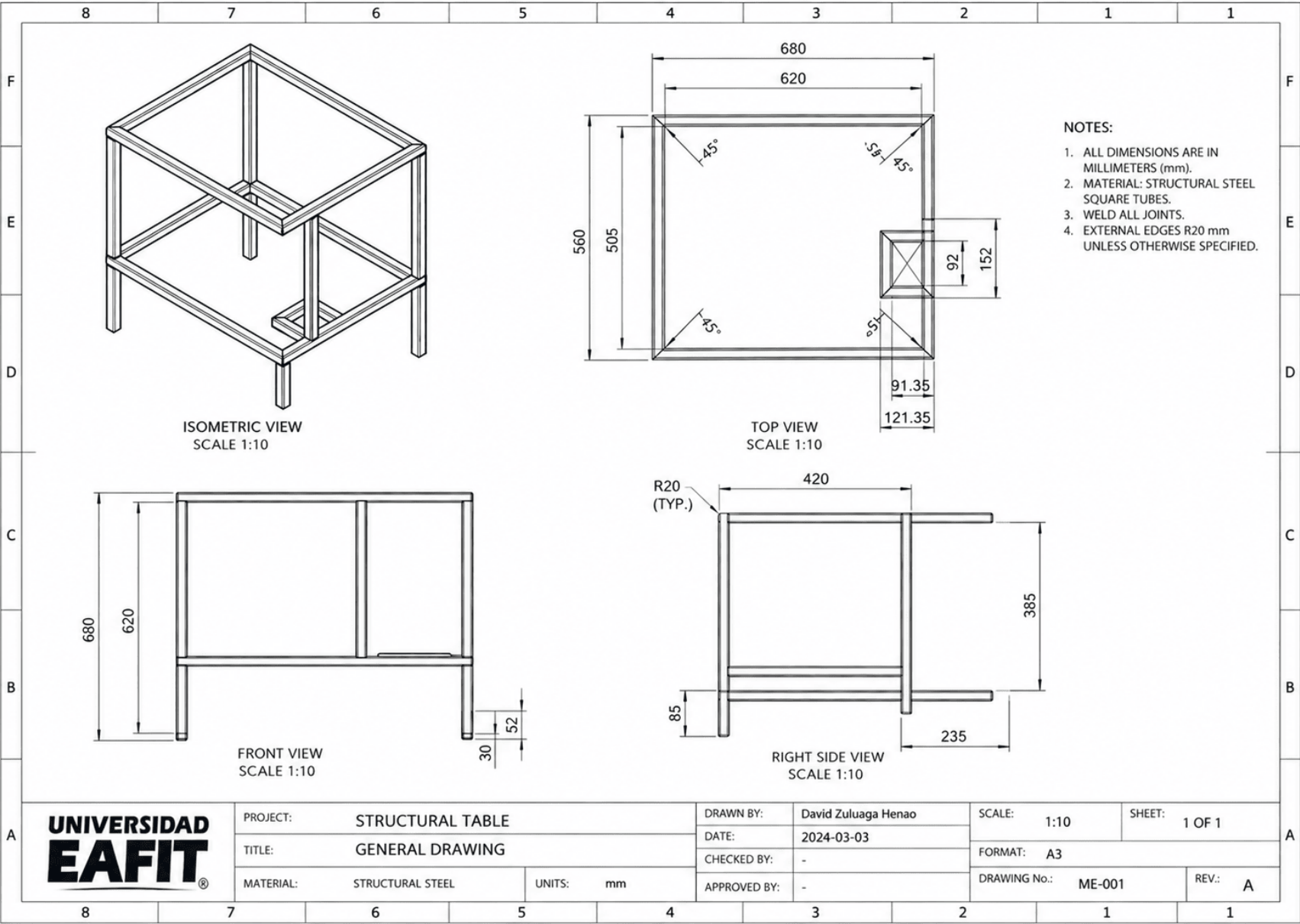
The “drawn by” column reproduces the title block of each CAD sheet. Sebastián Rincón Martínez carried the static analysis of Delivery 3 together with Juan Manuel Rodríguez Duque, and also contributed to the concept selection of Delivery 1.

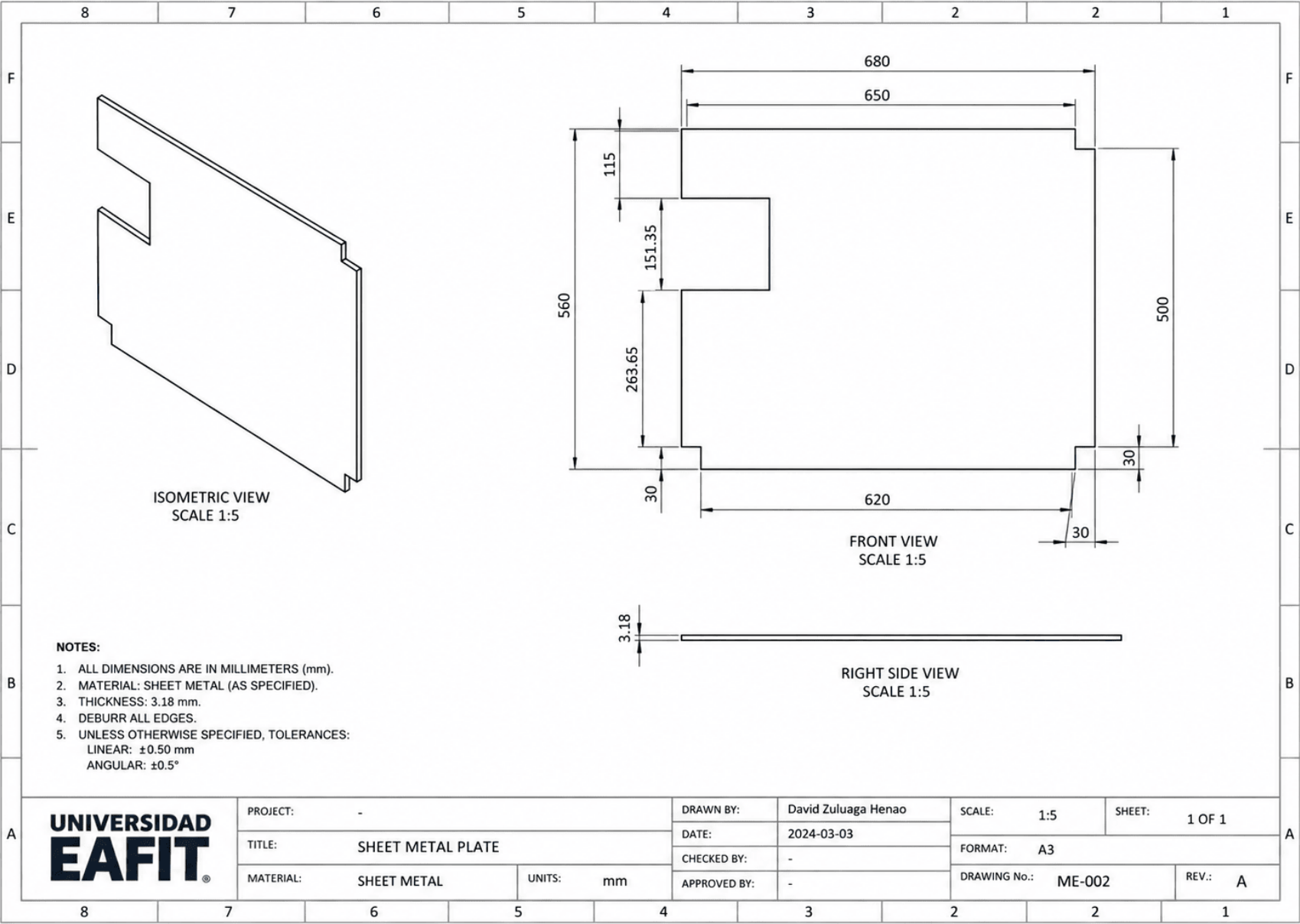
3. Relation to the other deliveries

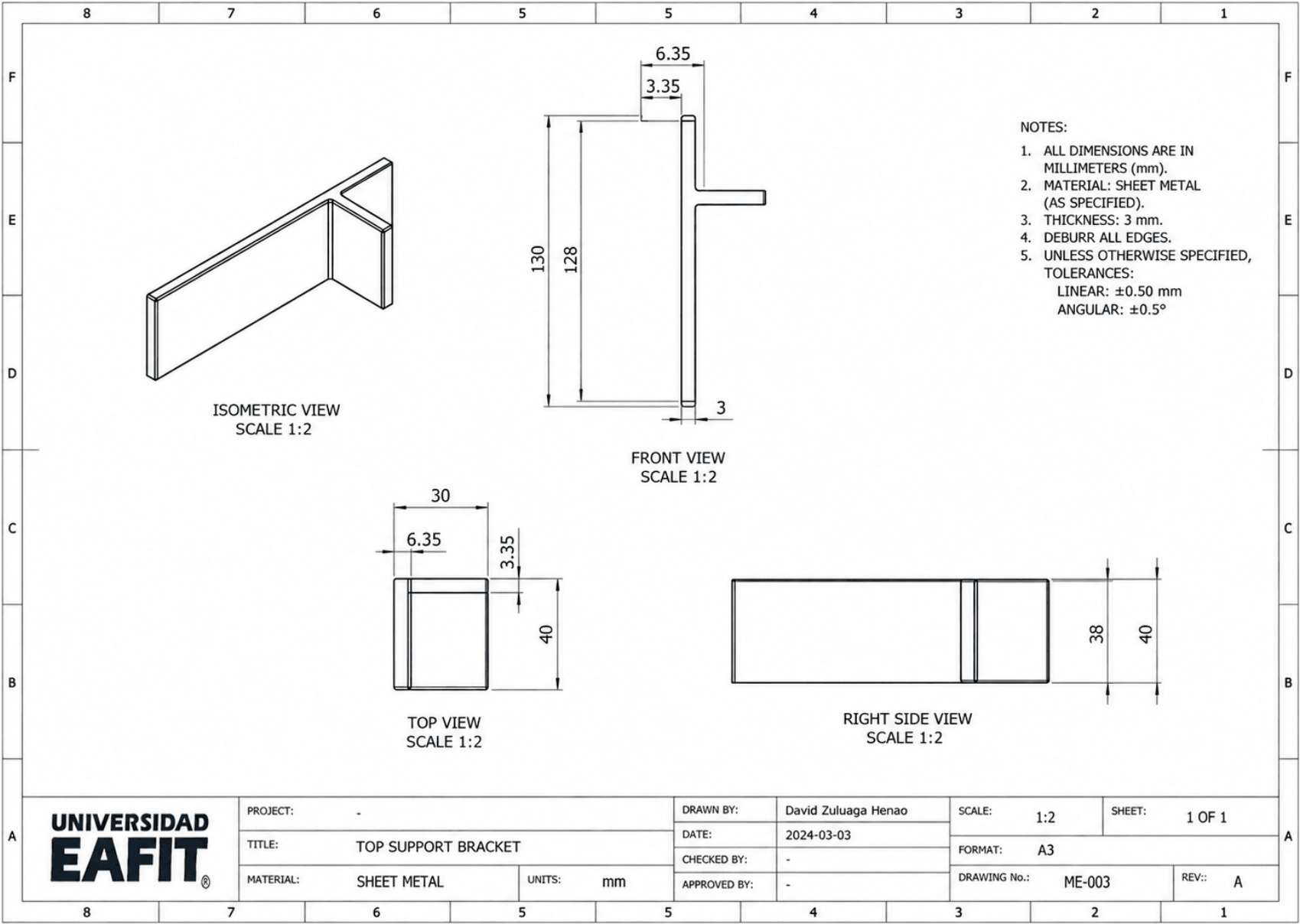
Group	Where it is verified
Chassis and structure	Delivery 3, Sheet 5: finite element analysis of the welded frame (safety factor 66, maximum deflection 0.102 mm).
Guide-supported axes	Delivery 3, Sheet 3: lead-screw torque, efficiency, stresses and column buckling (safety factor 3.21 against buckling), and Sheet 4: bushing loads and rod deflection of the carriages (safety factor 1.87, the tightest in the machine).
Arms	Delivery 3, Sheet 2: internal loads and bending stress in the six links (maximum 26.1 MPa, safety factor 9.6).
Assembled machine	Delivery 3, Sheet 1: kinematic model, stroke usage and workspace (a 200 mm cube, 8.6 dm ³).
Electrical system	Delivery 1: motor selection (NEMA 34 86HB250, 4.5 N m); Delivery 3, Sheet 3: required torque is 7.1 % of that.

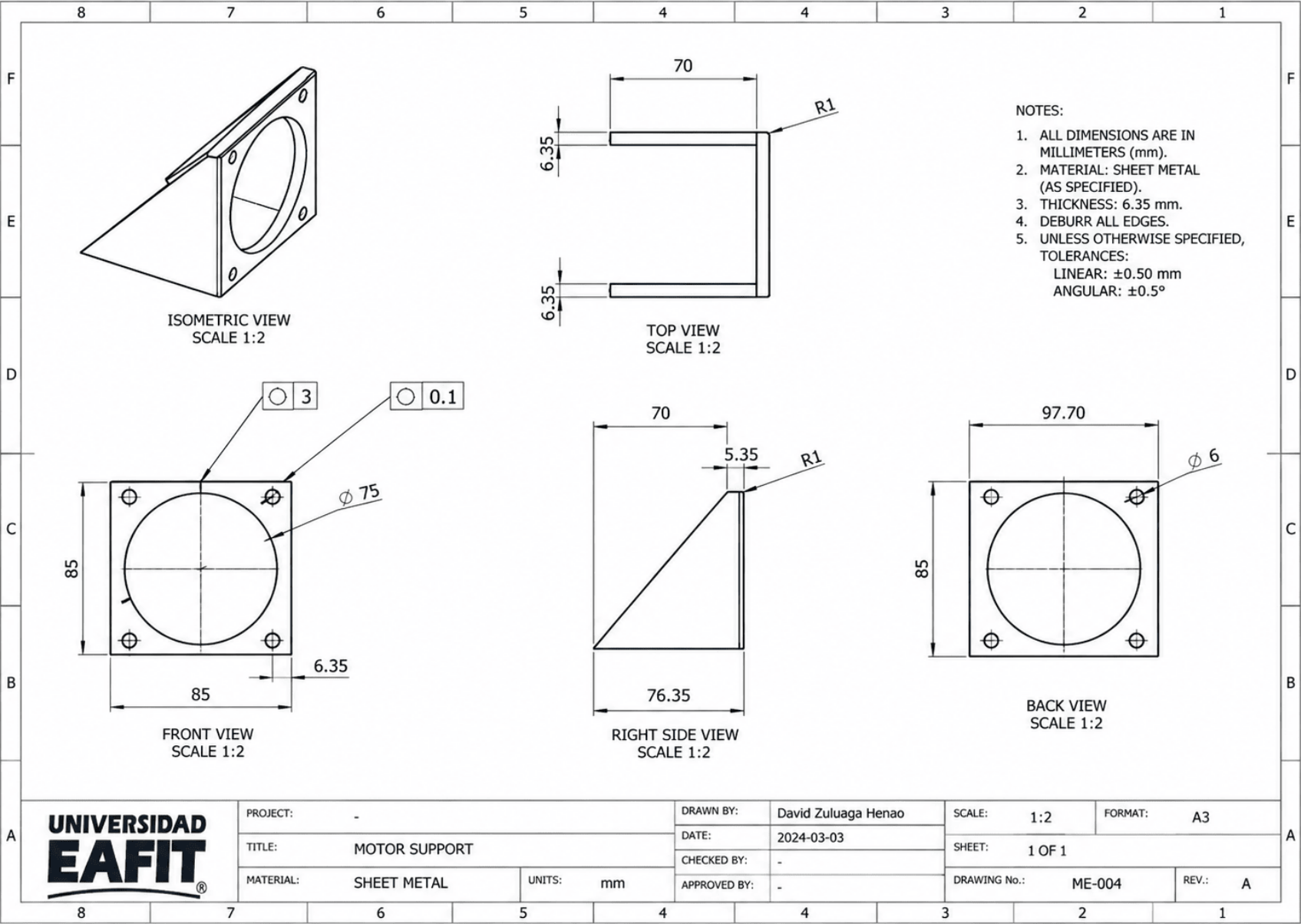
Note on the sheets that follow. The sheets marked *as issued* above keep their original Spanish title blocks. The reading key is: *Materia* = course, *Proyecto* = project, *Dibujó* = drawn by, *Verif.* = checked, *Aprob.* = approved, *Escala* = scale, *Unidades* = units, *Cantidad* = quantity, *Formato* = sheet size, *Material* = material, *Hoja 1 de 1* = sheet 1 of 1, *Las cotas se expresan en mm* = all dimensions in millimetres, *No cambie la escala* = do not scale the drawing, *Romper aristas* = break sharp edges.

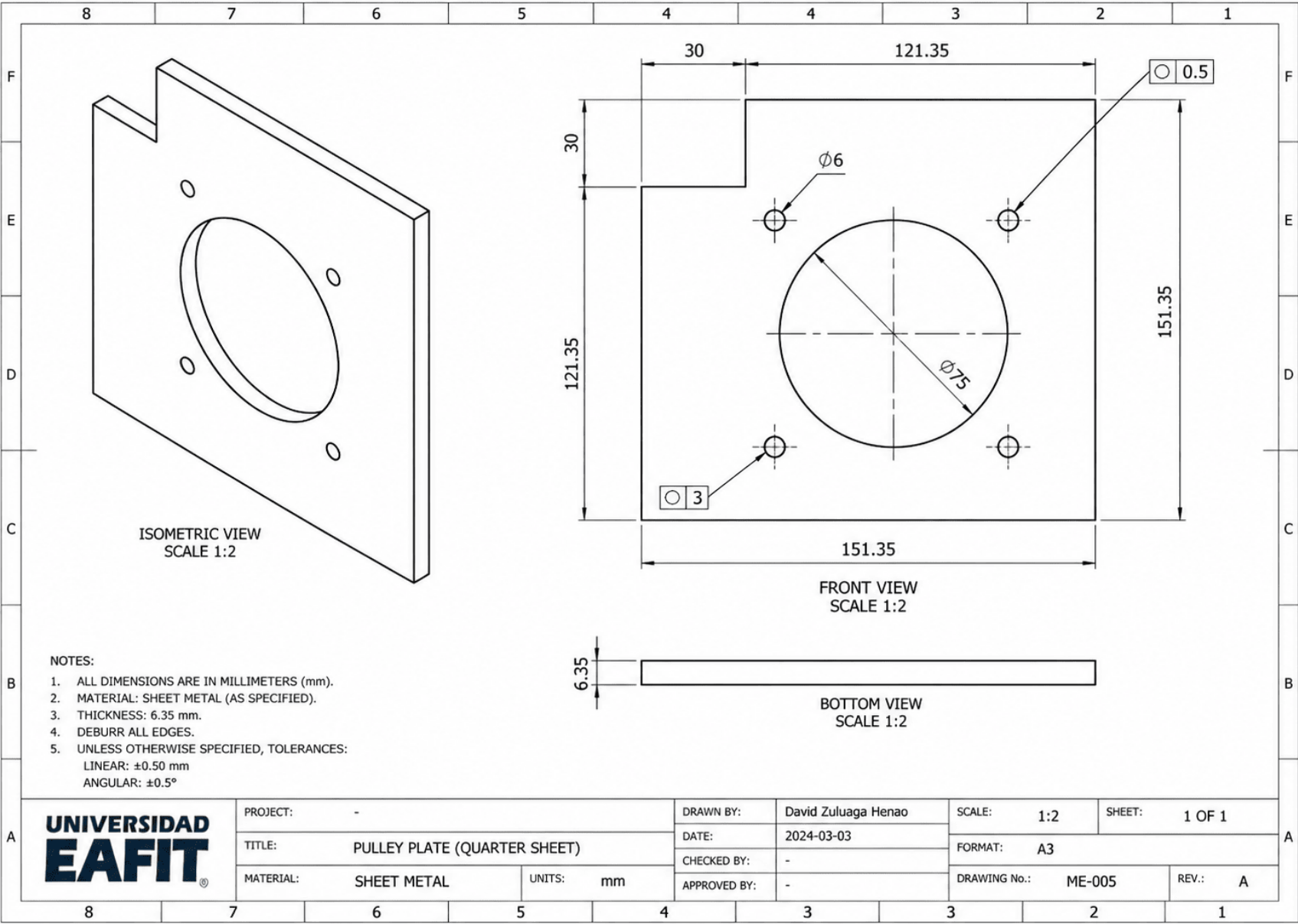
Group 1 – Chassis and structure

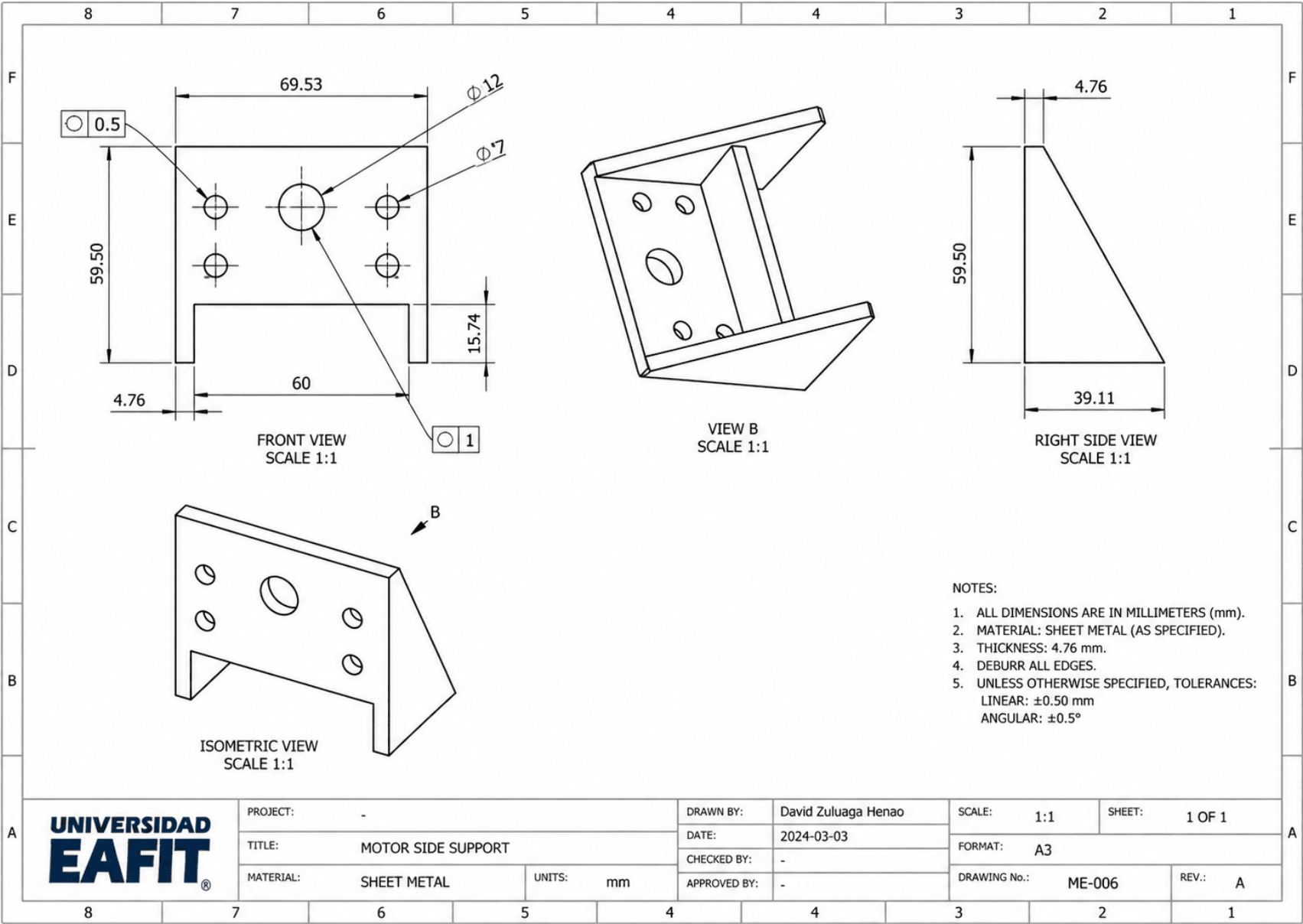


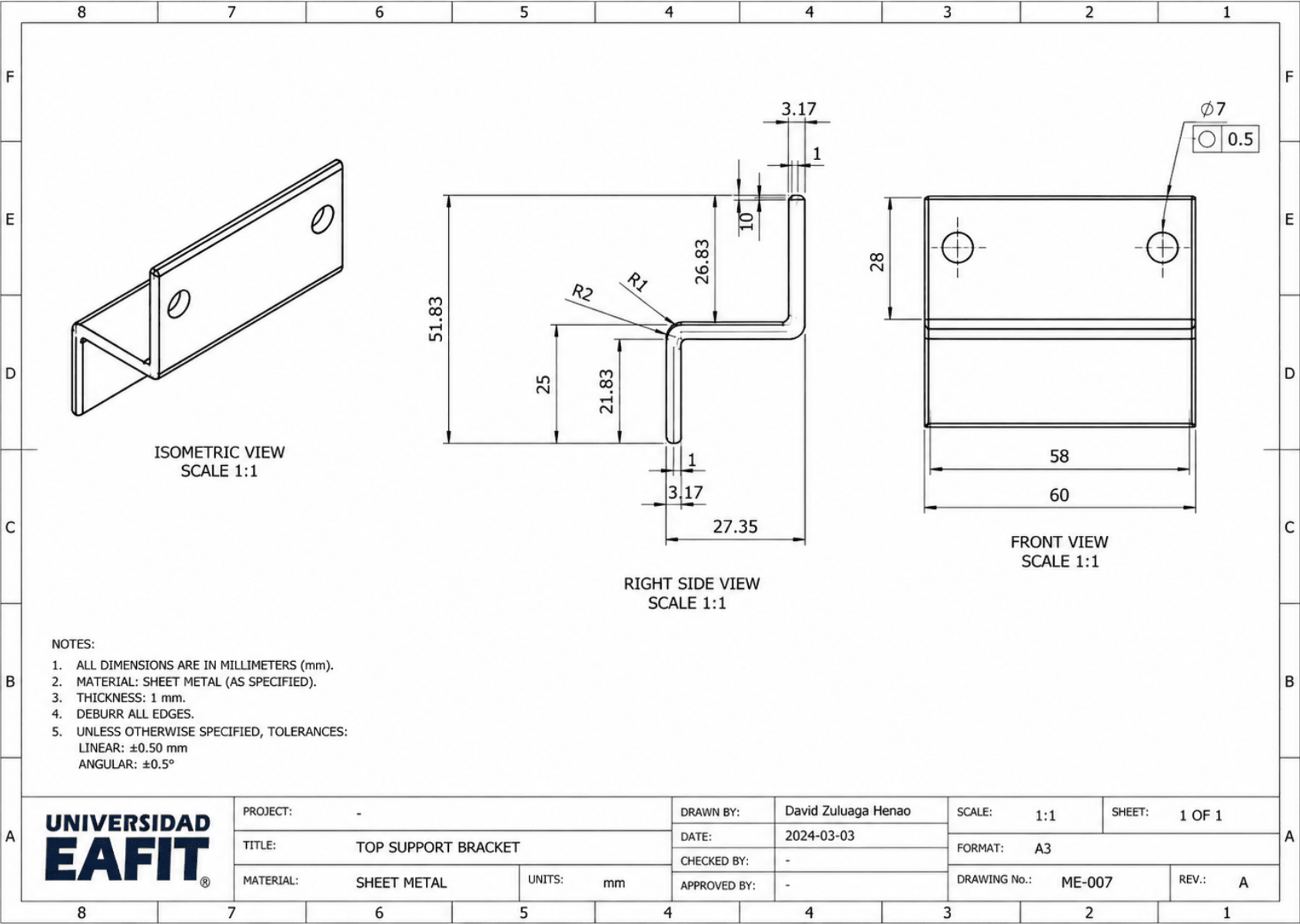


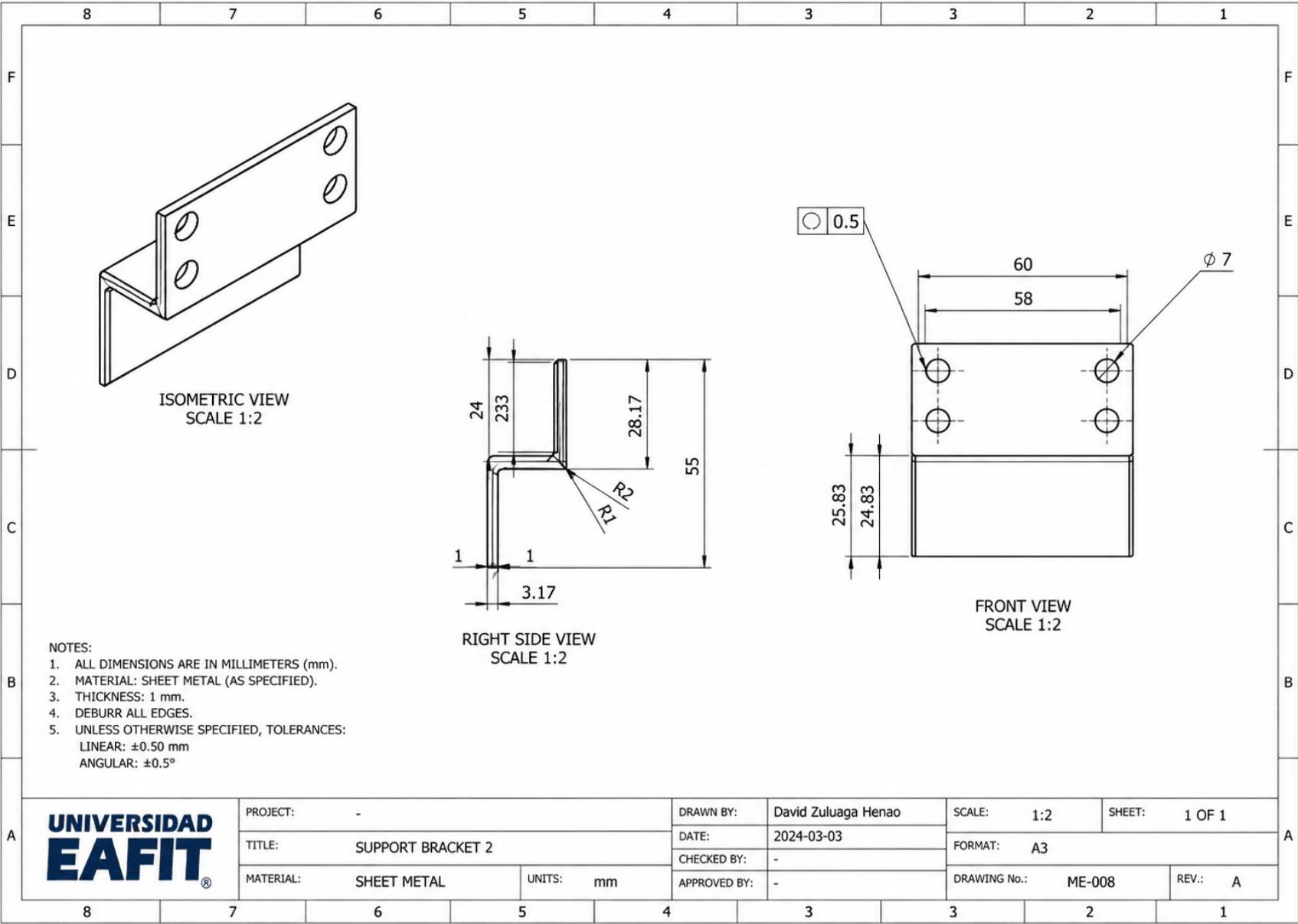




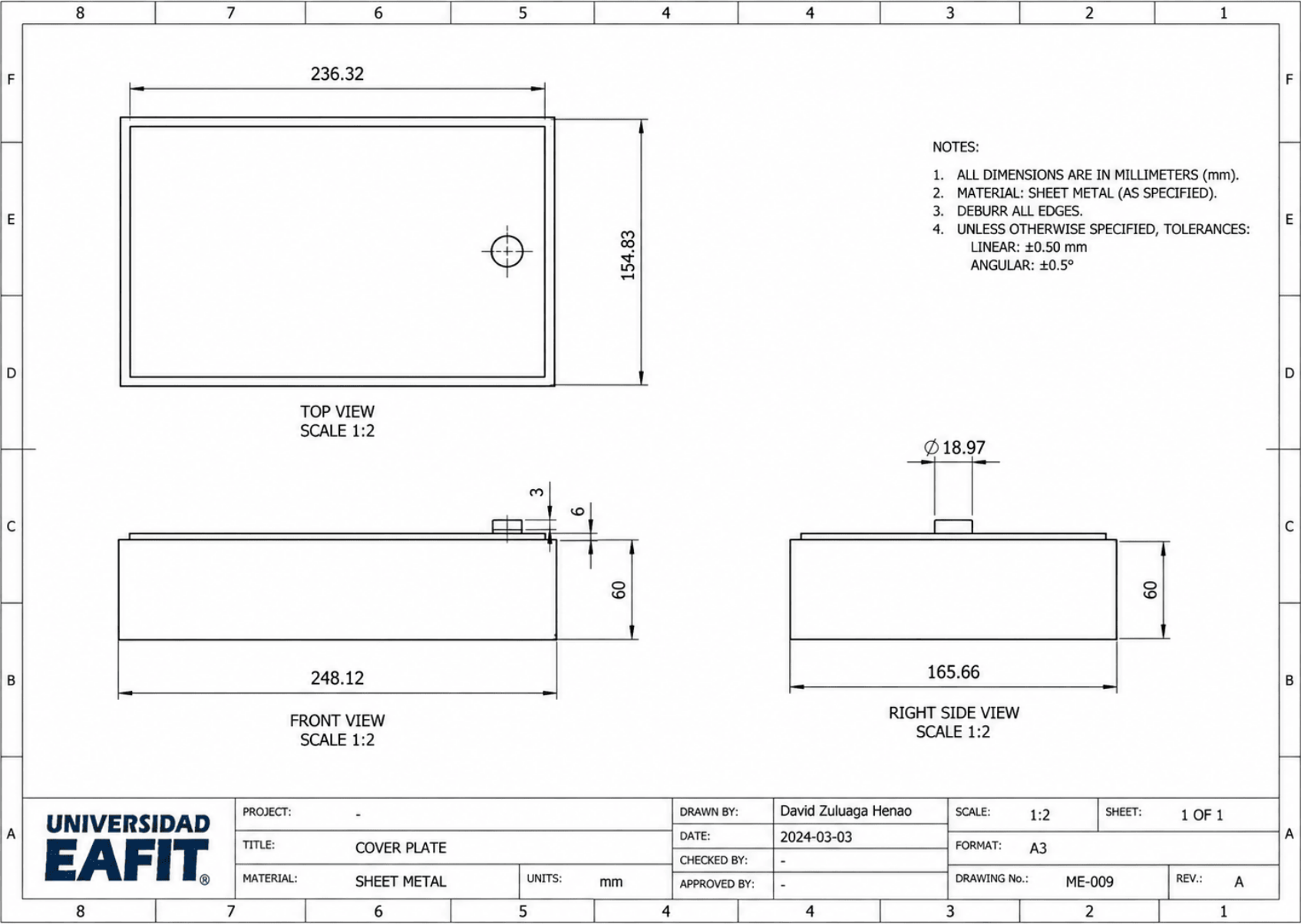


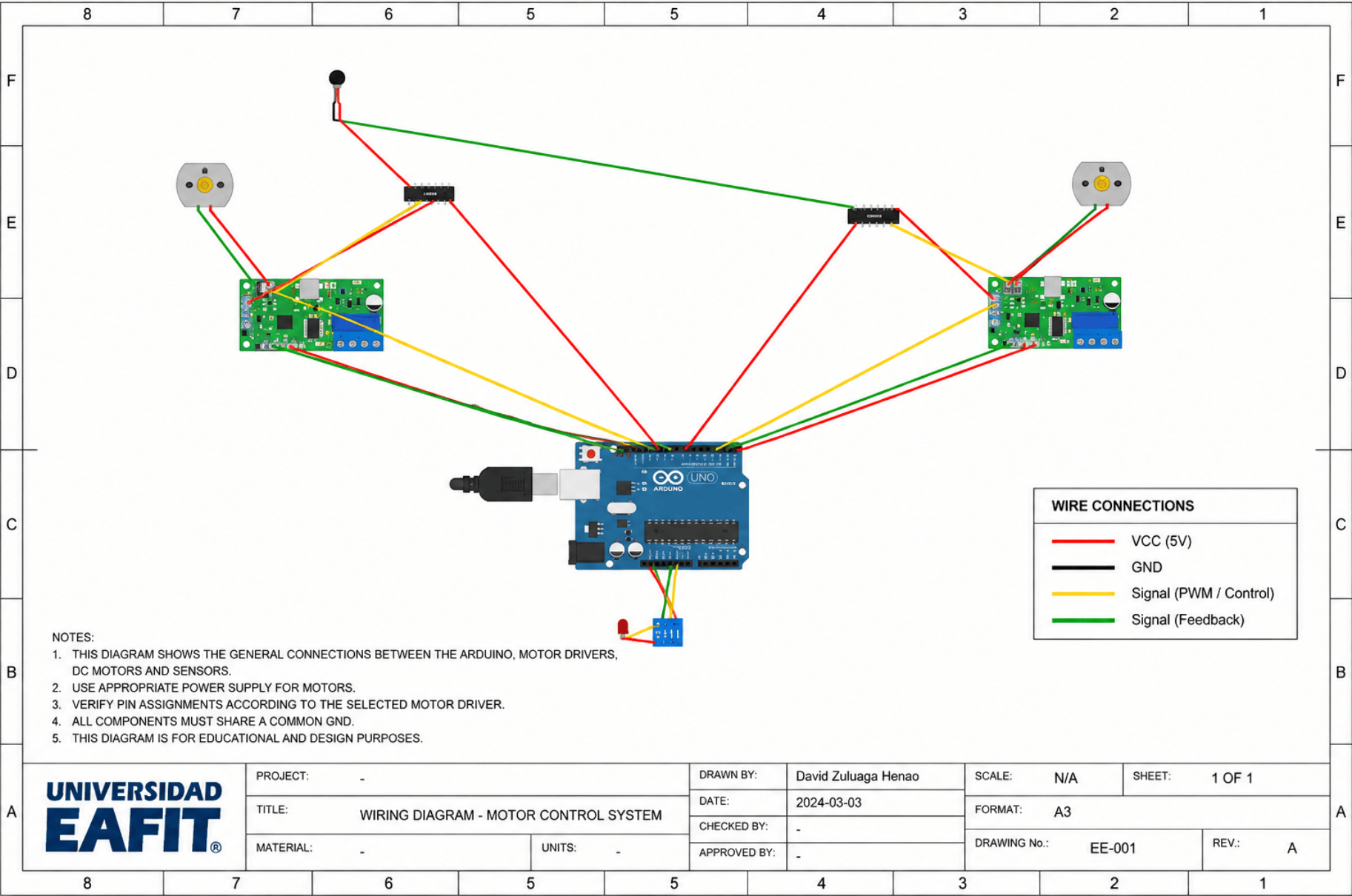


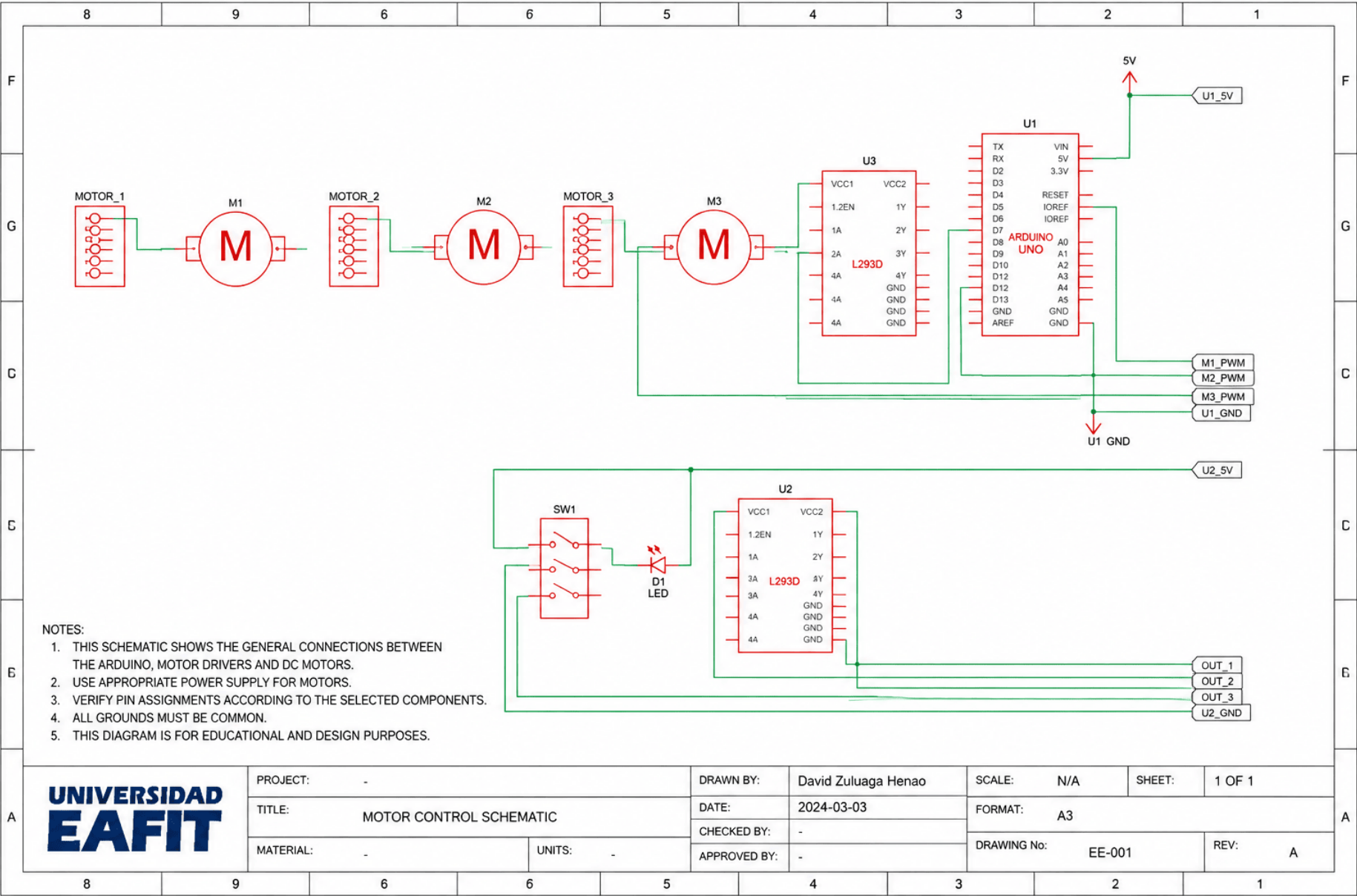




Group 2 – Electrical system







	8	7	6	5	4	4	3	2	1	
F										F
E										E
D										D
C										C
B										B
A										A
	8	7	6	5	4	4	3	2	1	

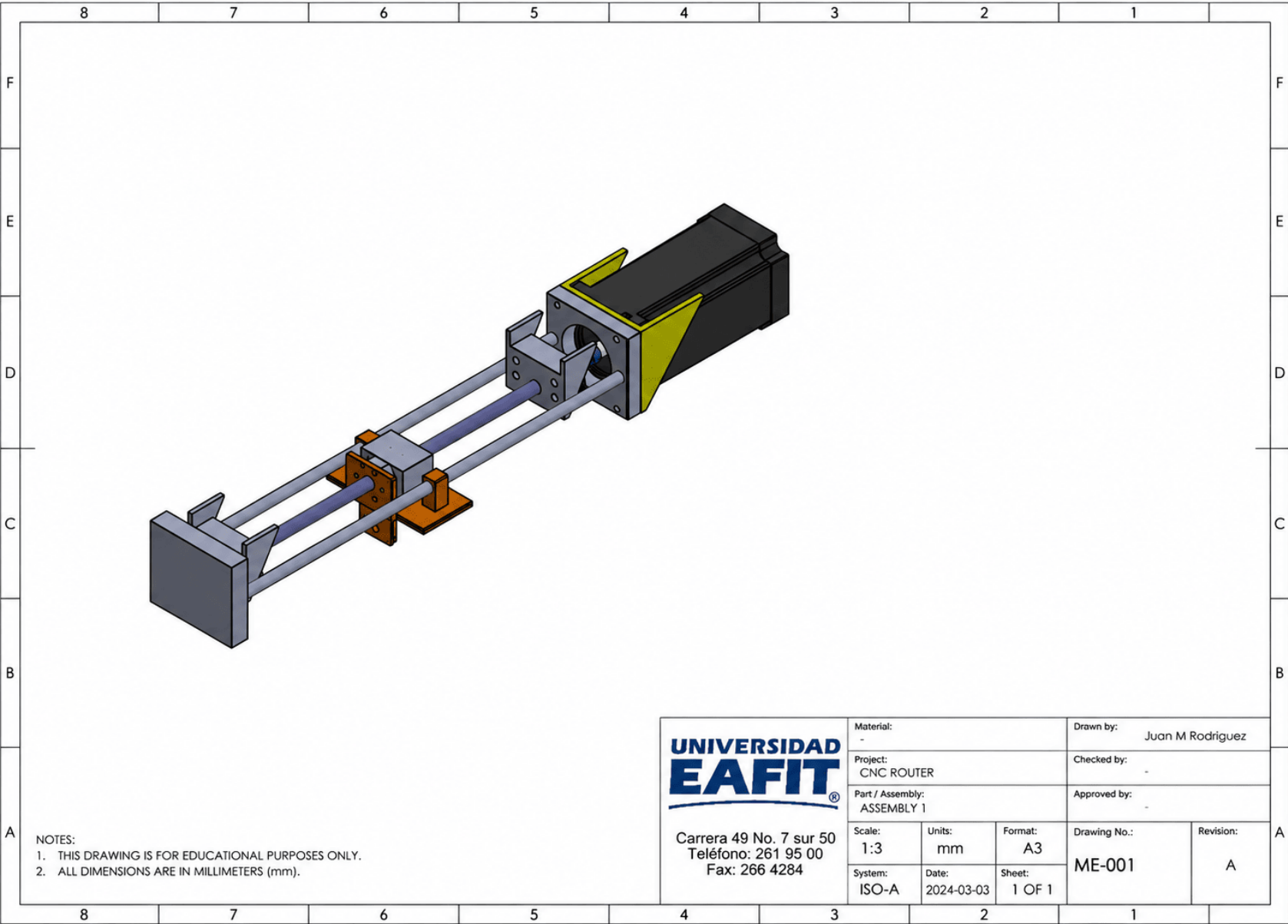
ID	Quantity	Component	Description / Specifications
M1, M2	2	DC Motor	Brushed DC motor
U1	1	Arduino Uno R3	Microcontroller board
POLOLU1, POLOLU2	2	Pololu Motor Driver (Single)	Simple motor controller (e.g., Pololu DRV8833)
U2, U3	2	H-Bridge Motor Driver	Bridge motor controller (e.g., L293D)
SW1	1	DIP Switch	SPST DIP switch x4
D1	1	Red LED	Indicator LED (Red)
M3	1	Vibration Motor	DC vibration motor

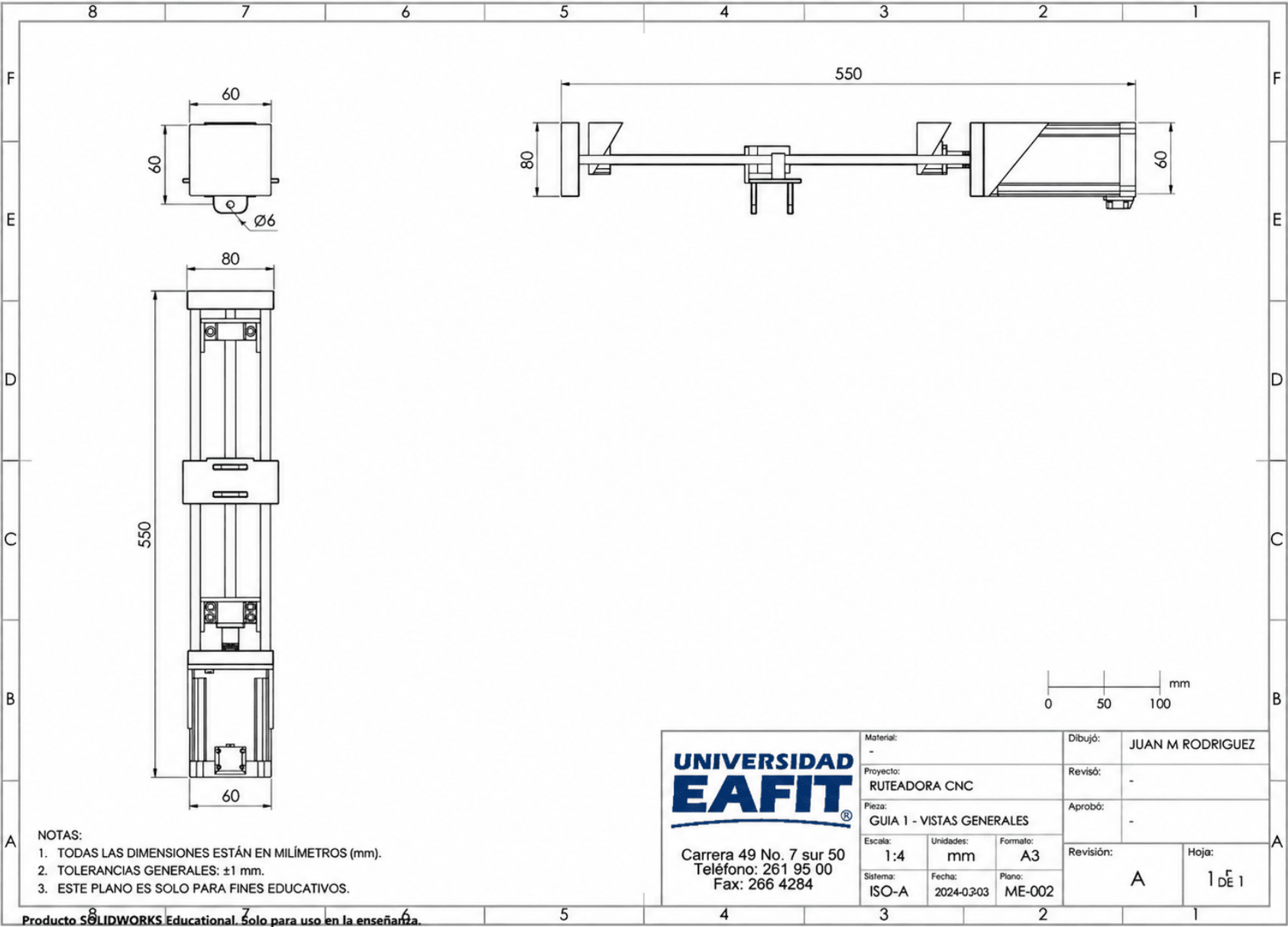
NOTES:

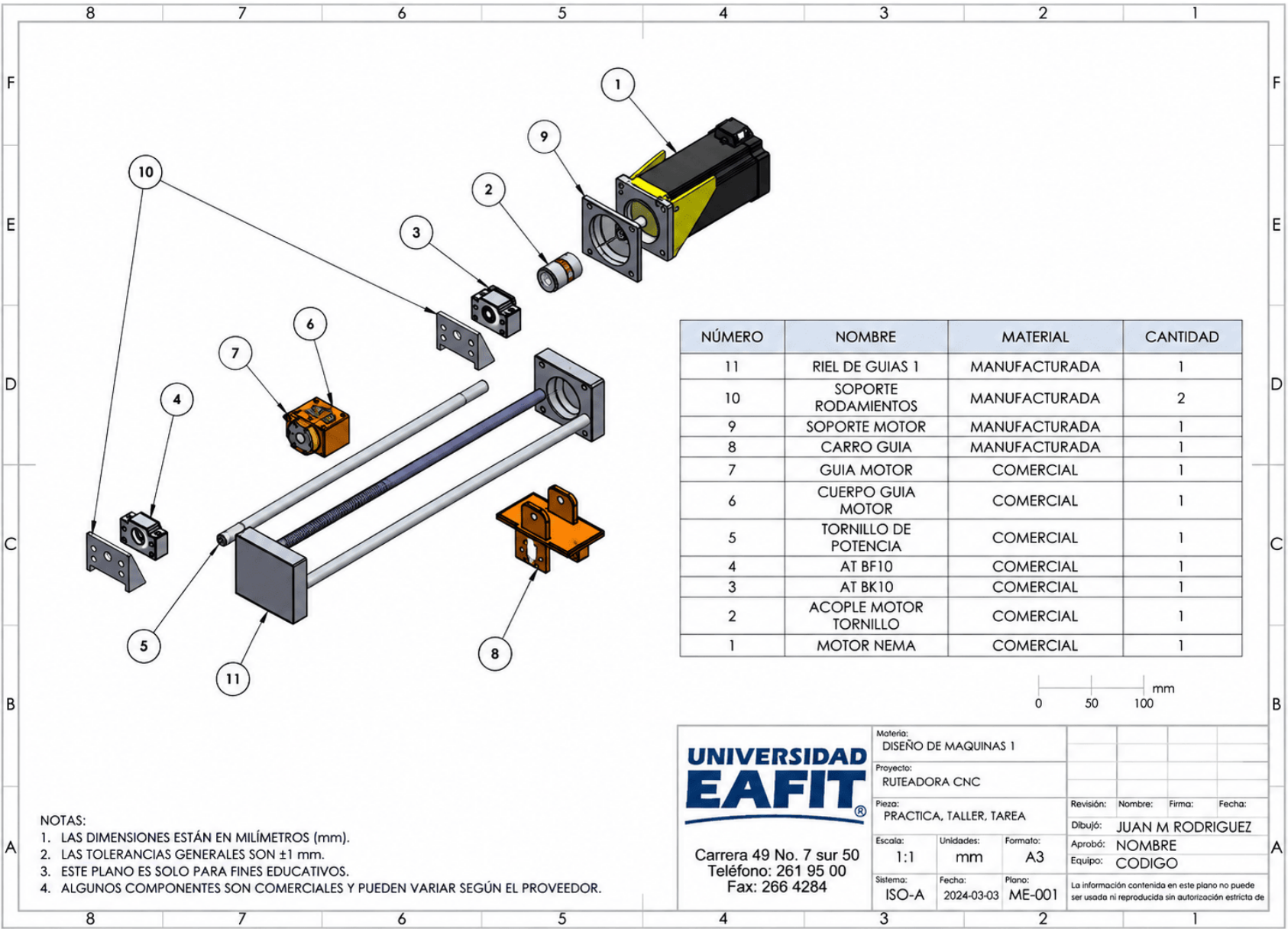
1. QUANTITIES ARE FOR ONE COMPLETE SYSTEM.
2. COMPONENT SPECIFICATIONS MAY VARY ACCORDING TO AVAILABILITY.
3. USE COMPATIBLE POWER SUPPLY FOR MOTORS AND DRIVERS.
4. THIS LIST IS FOR EDUCATIONAL AND DESIGN PURPOSES.

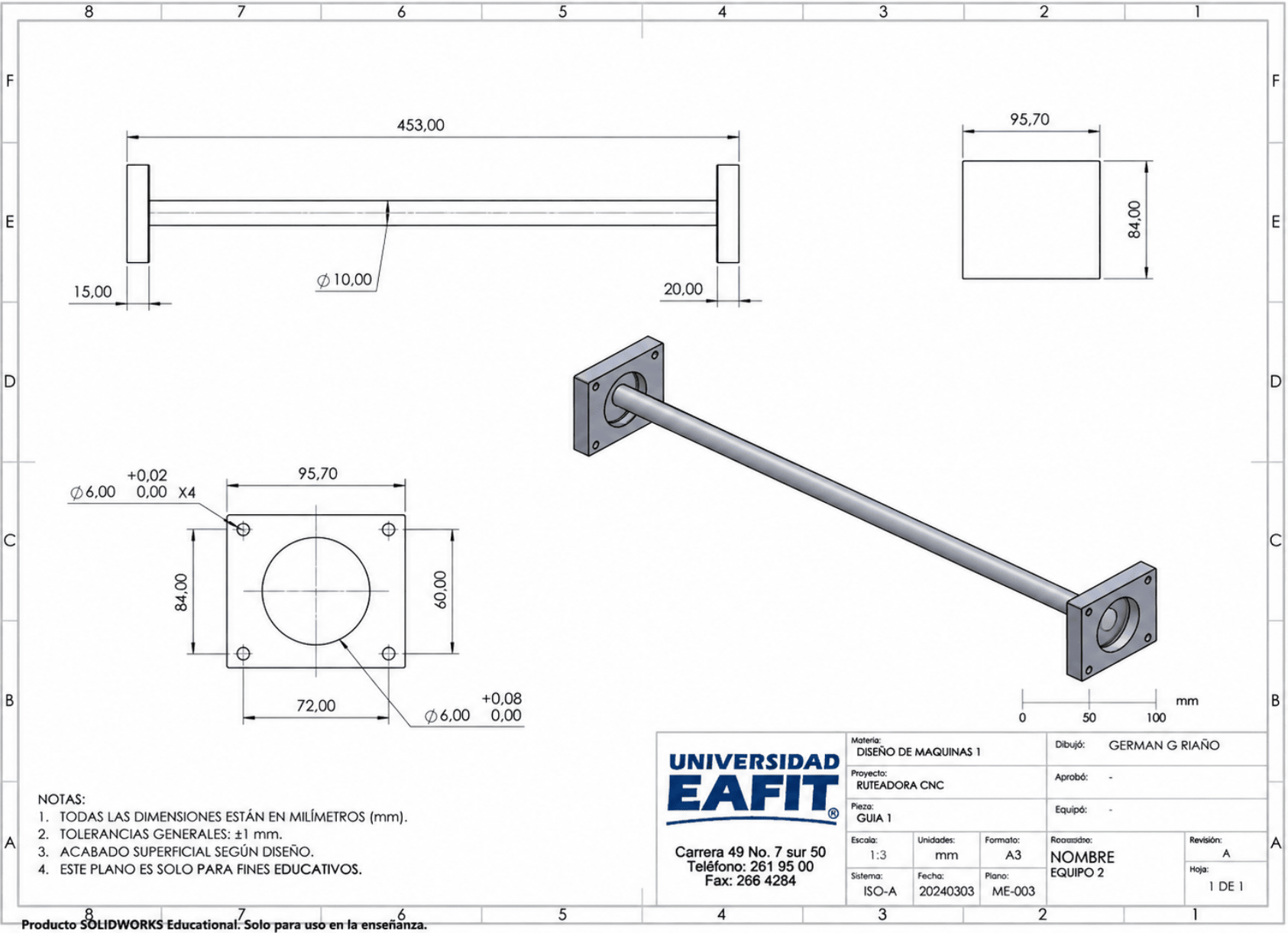
	PROJECT:	-	DRAWN BY:	David Zuluaga Henao	SCALE:	N/A	SHEET:	1 OF 1
	TITLE:	BILL OF MATERIALS	DATE:	2024-03-03	FORMAT:	A3		
	MATERIAL:	-	UNITS:	-	CHECKED BY:	-	DRAWING No.:	EE-002
					APPROVED BY:	-	REV.:	A

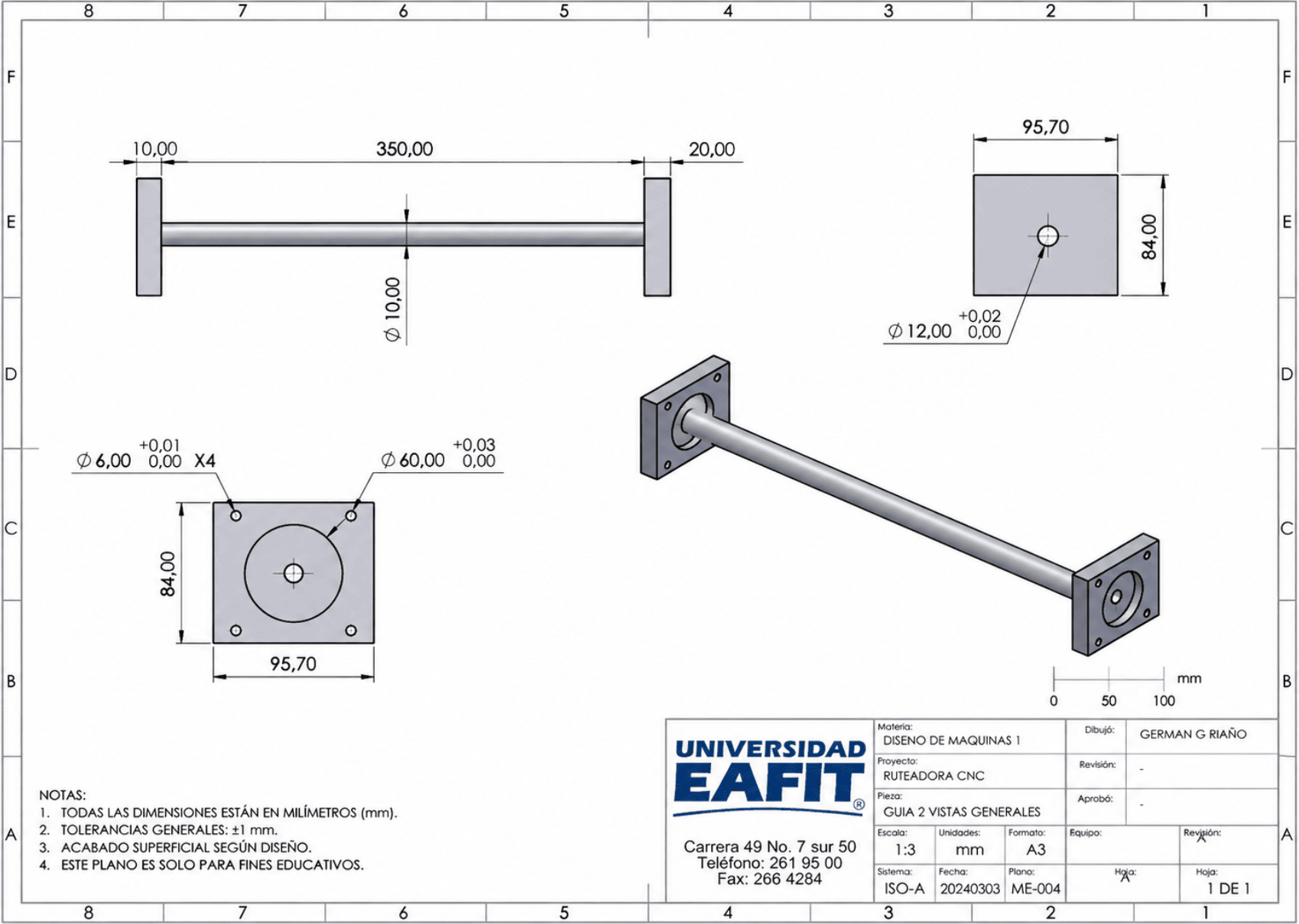
Group 3 – Guide-supported axis system

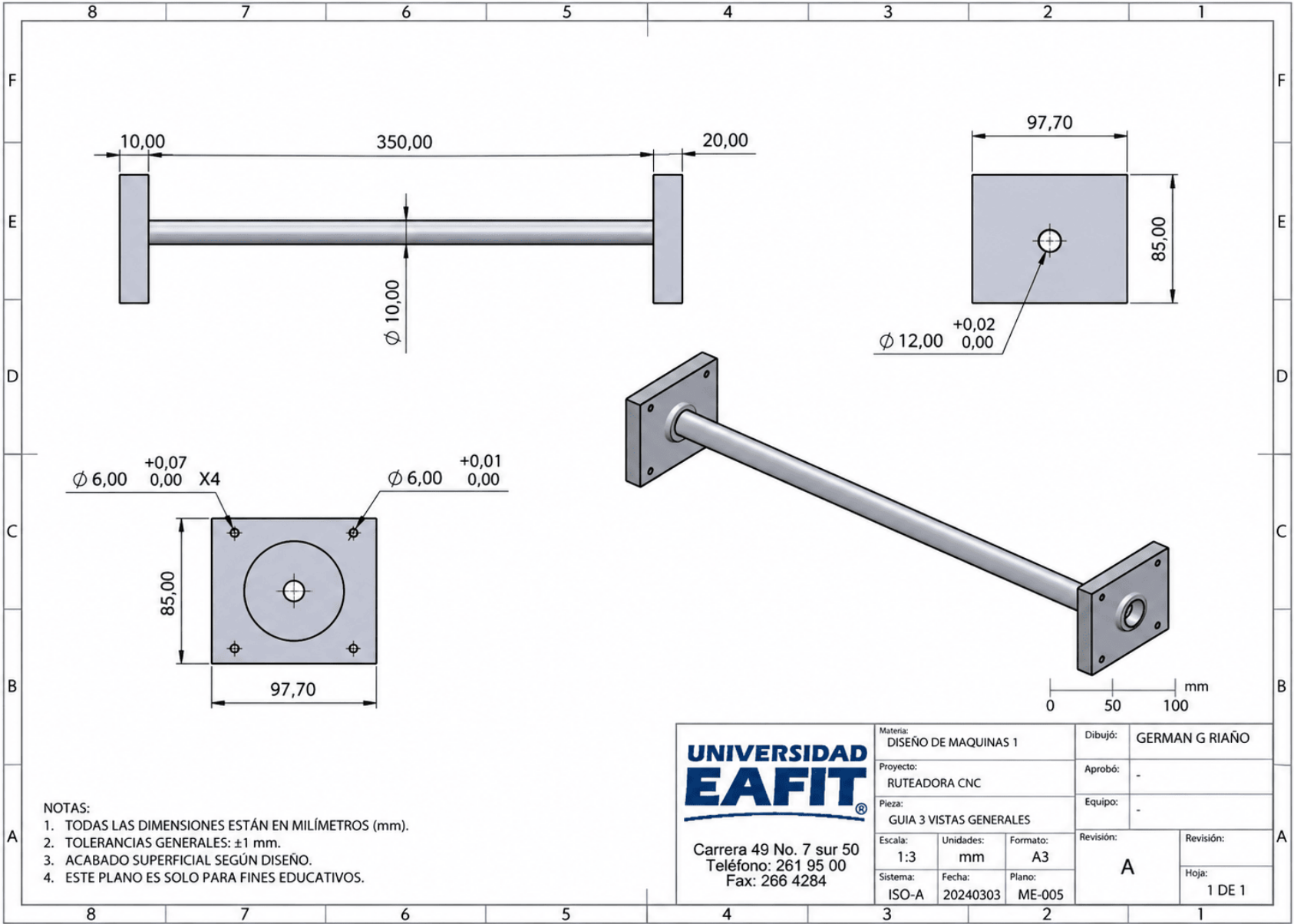


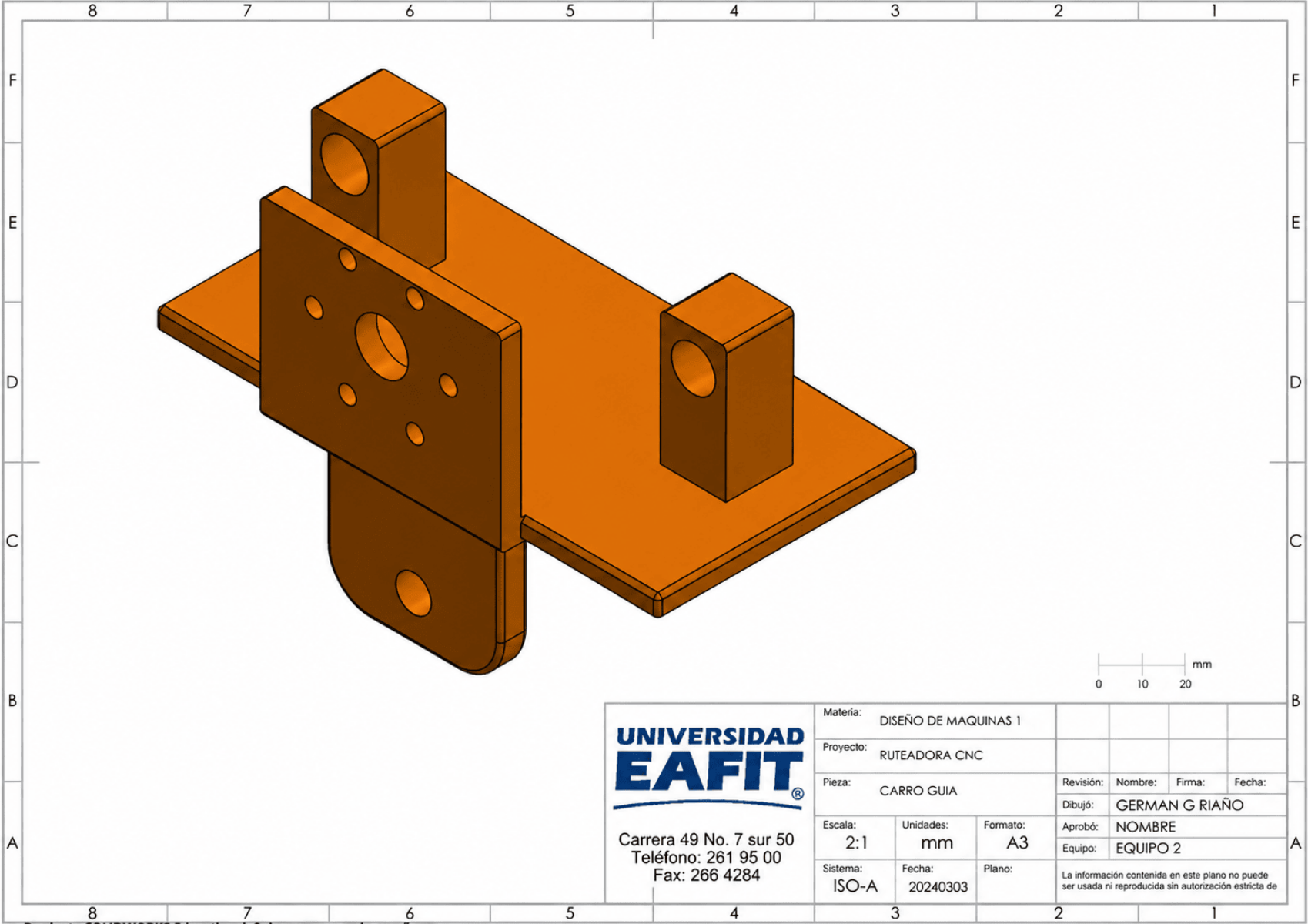






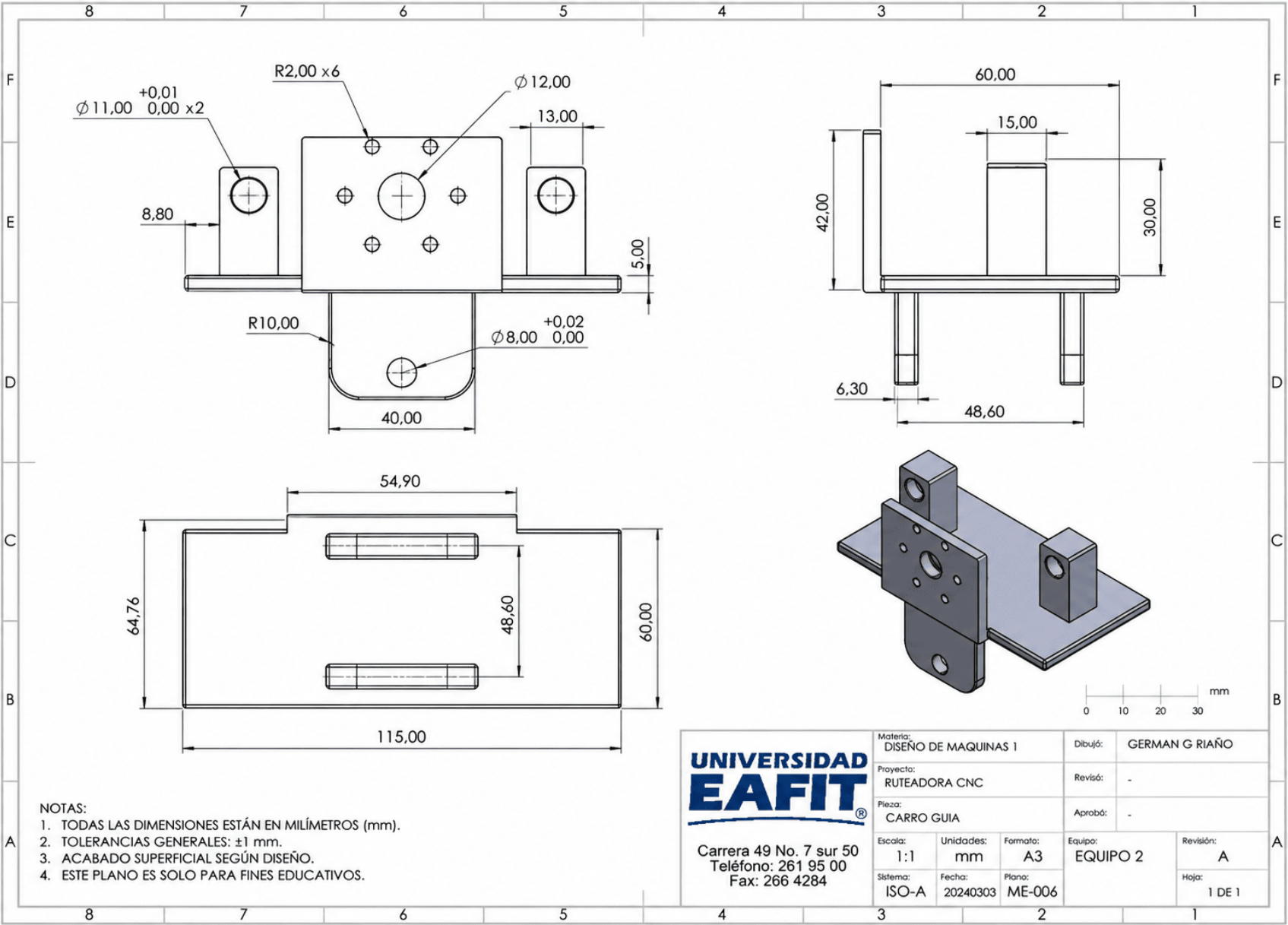


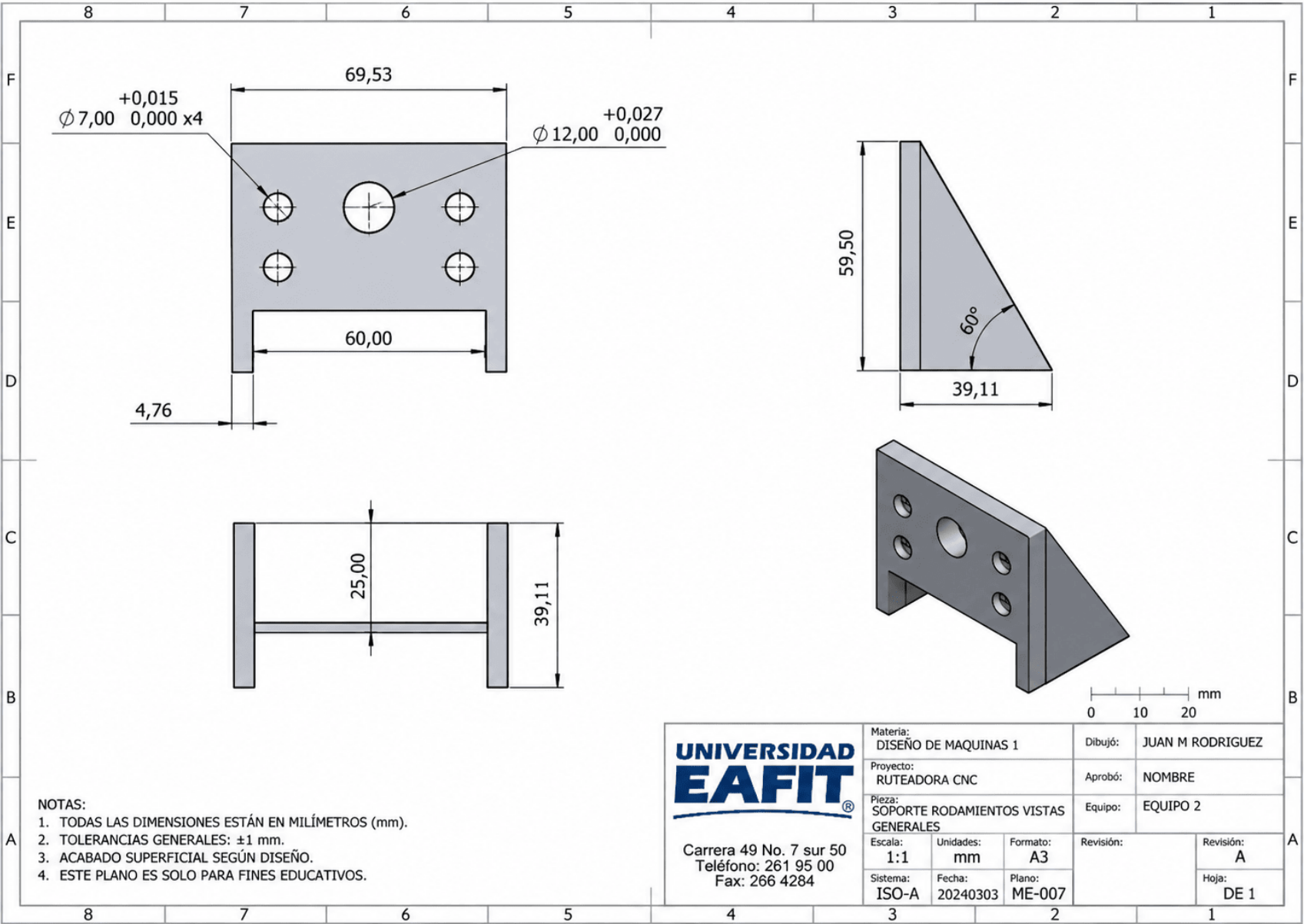




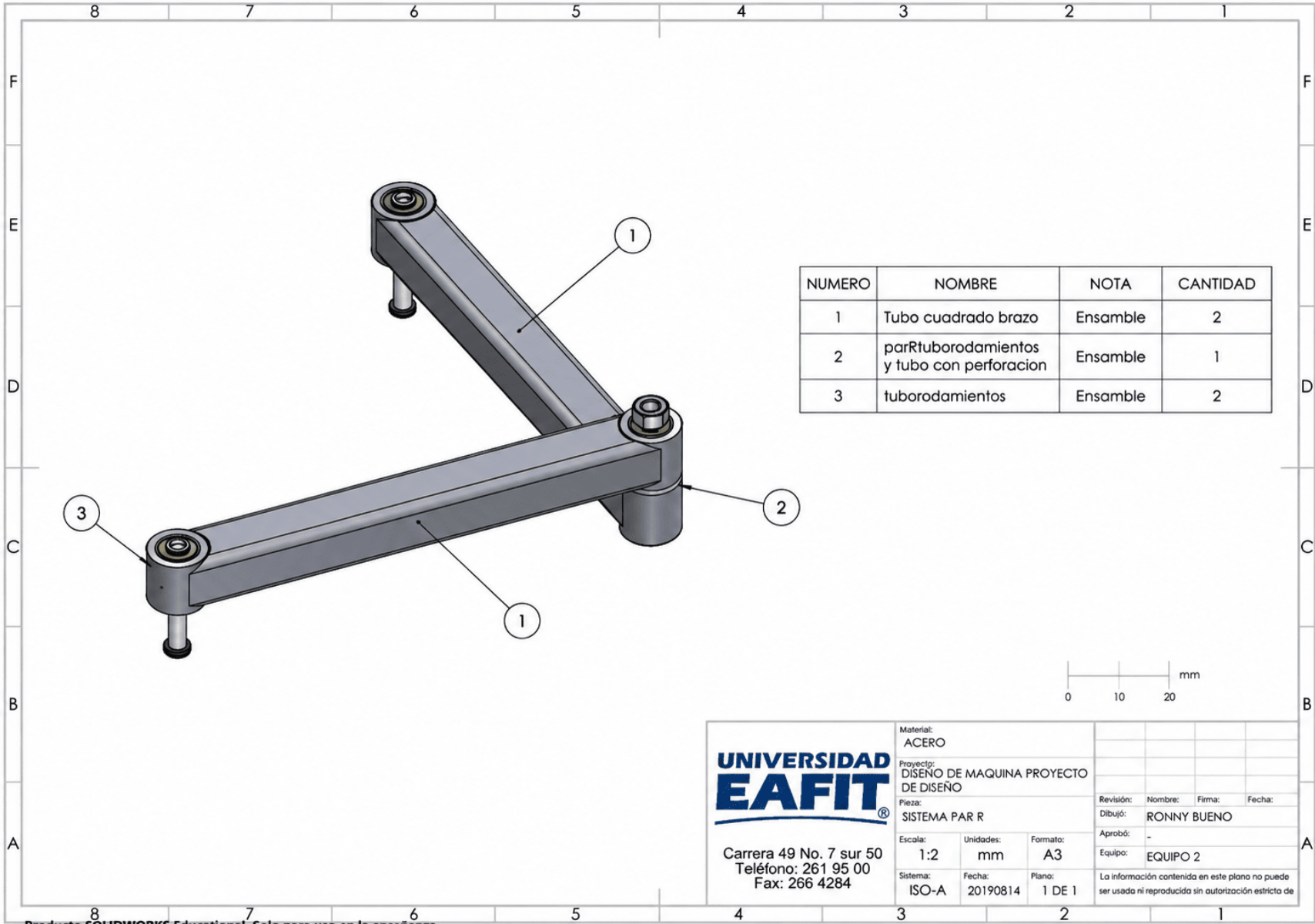
Carrera 49 No. 7 sur 50
 Teléfono: 261 95 00
 Fax: 266 4284

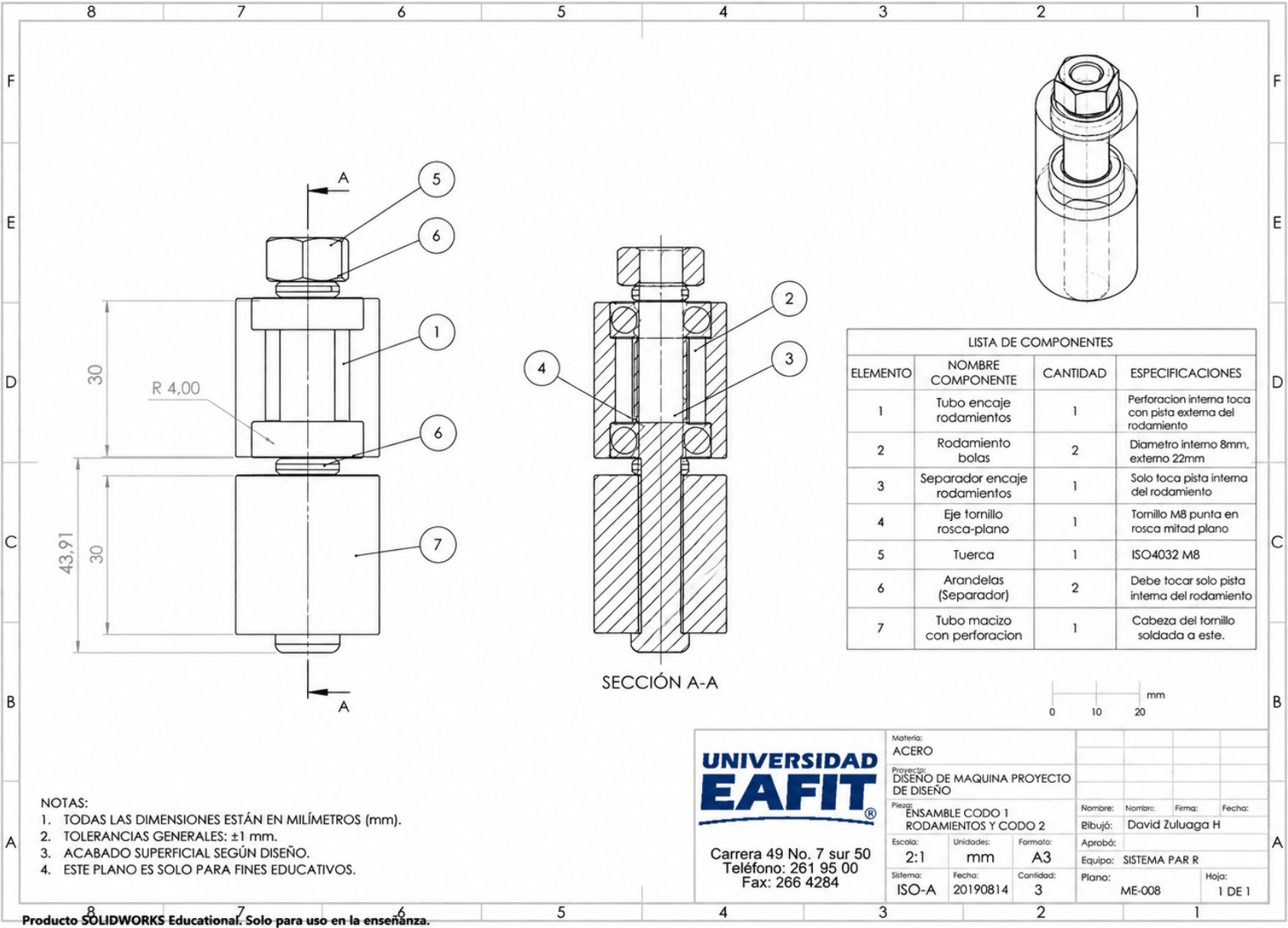
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Proyecto: RUTEADORA CNC						
Pieza: CARRO GUIA			Revisión:	Nombre:	Firma:	Fecha:
Escala: 2:1			Dibujó: GERMAN G RIAÑO			
Unidades: mm			Aprobó: NOMBRE			
Formato: A3			Equipo: EQUIPO 2			
Sistema: ISO-A			Fecha: 20240303			
Plano:			La información contenida en este plano no puede ser usada ni reproducida sin autorización estricta de			

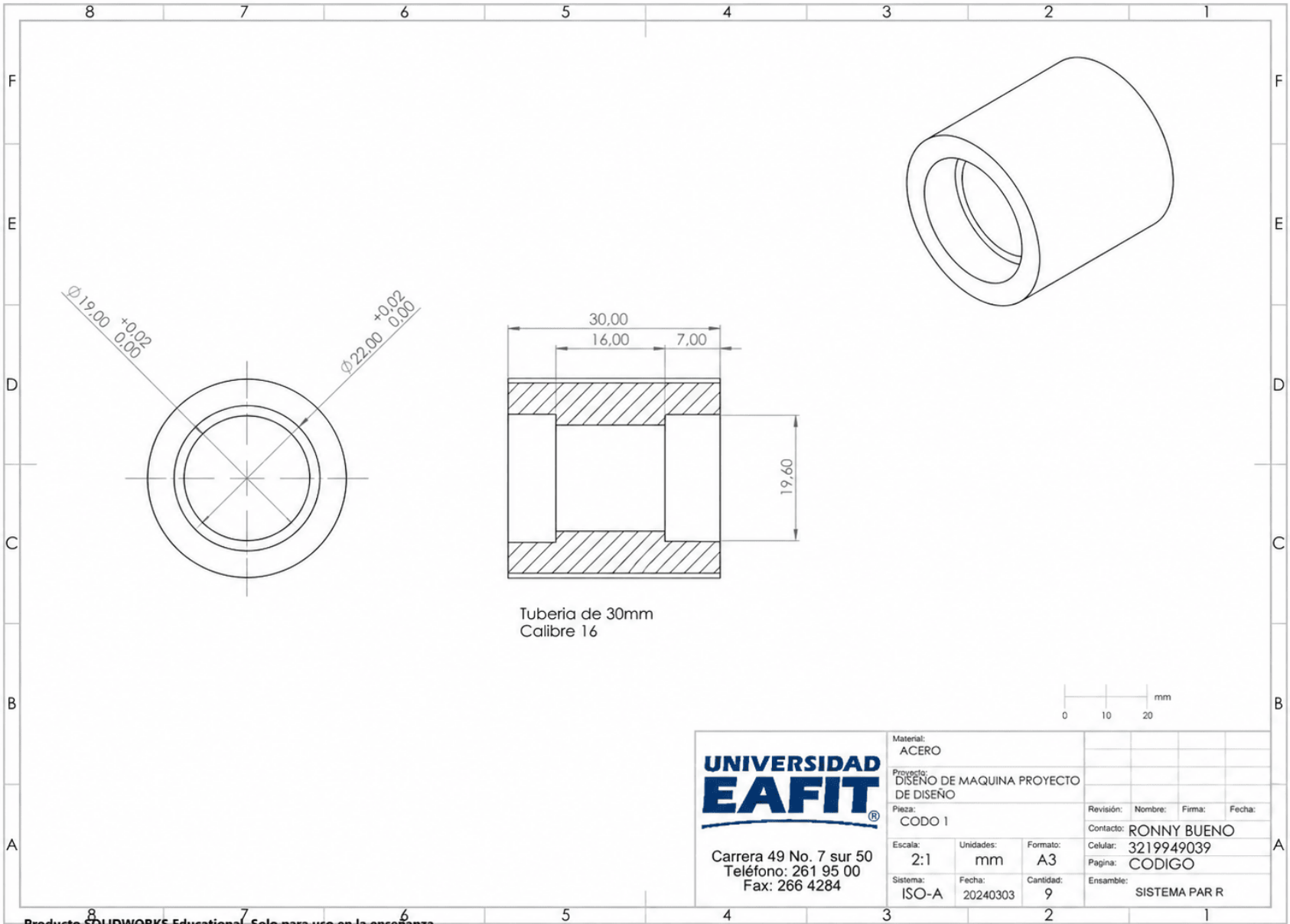


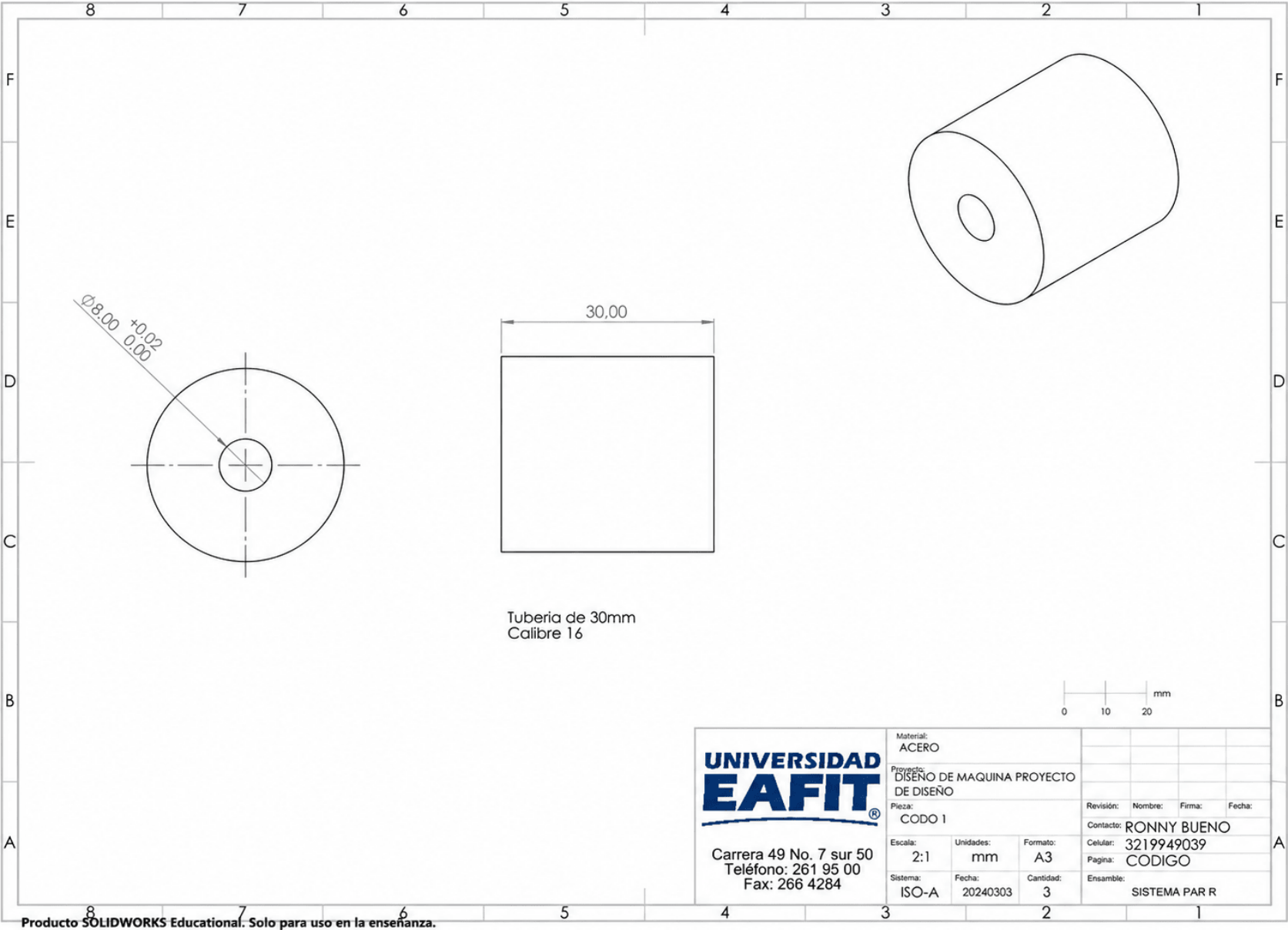


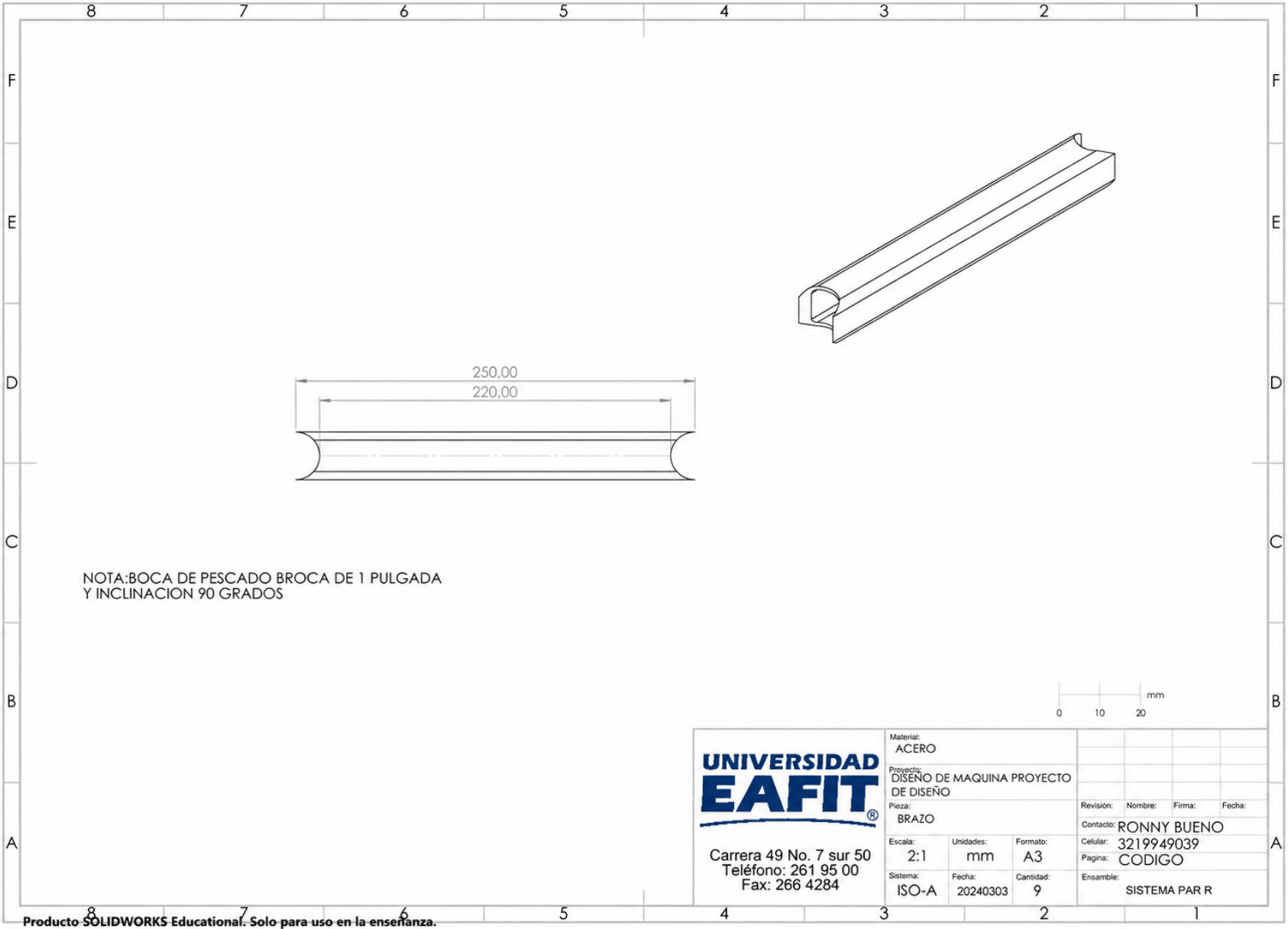
Group 4 – Arms (R-pair system)

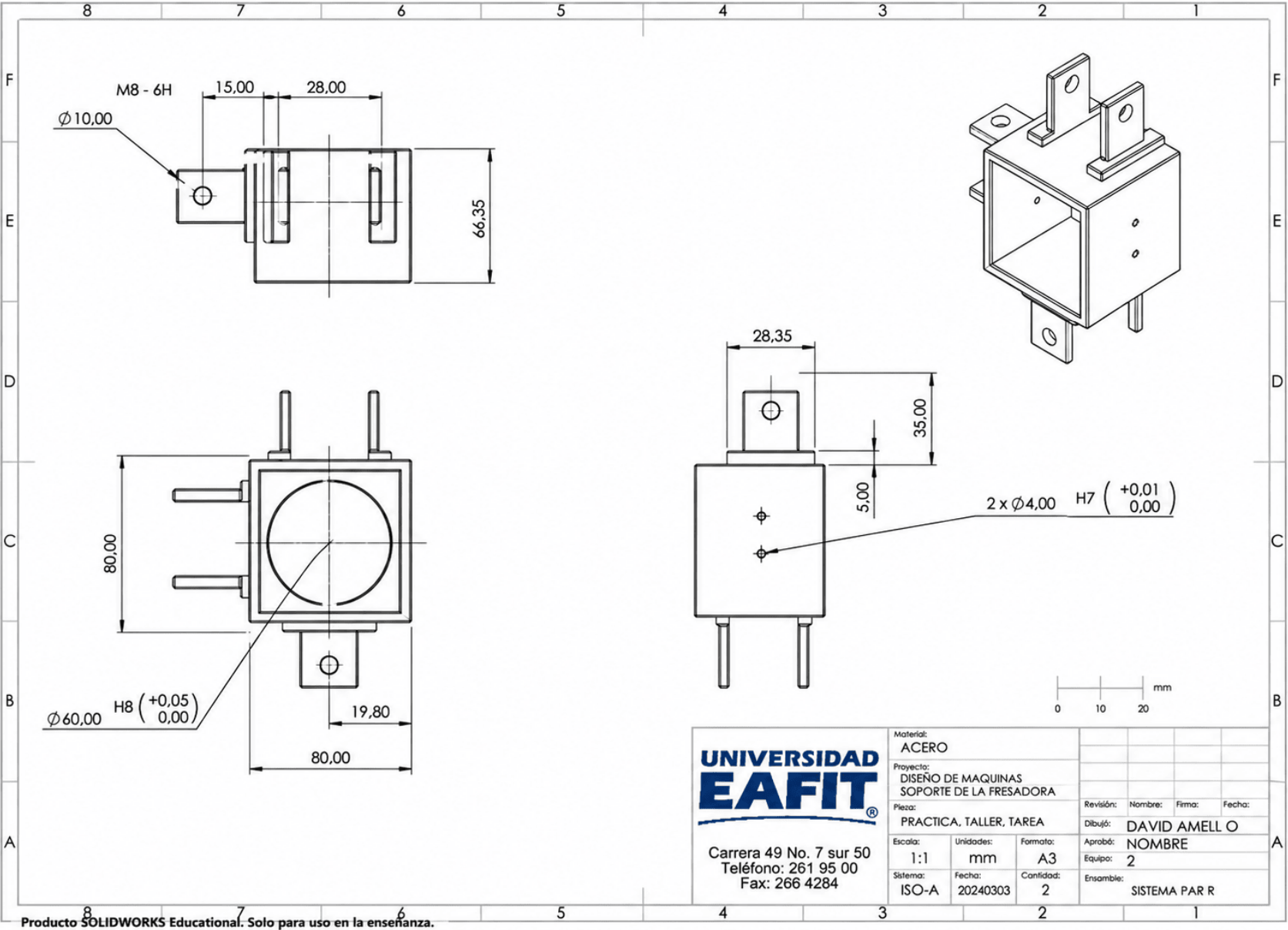


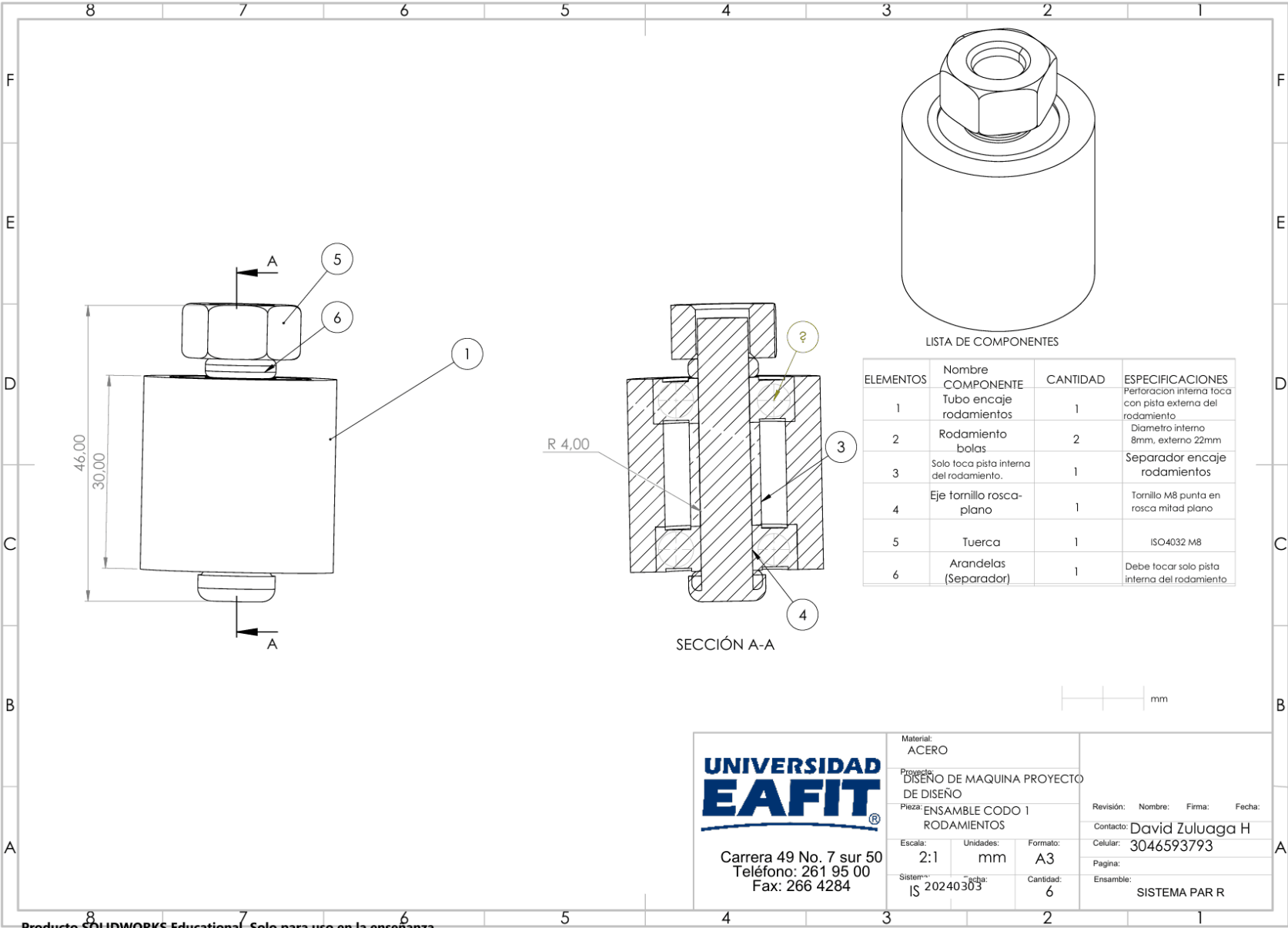




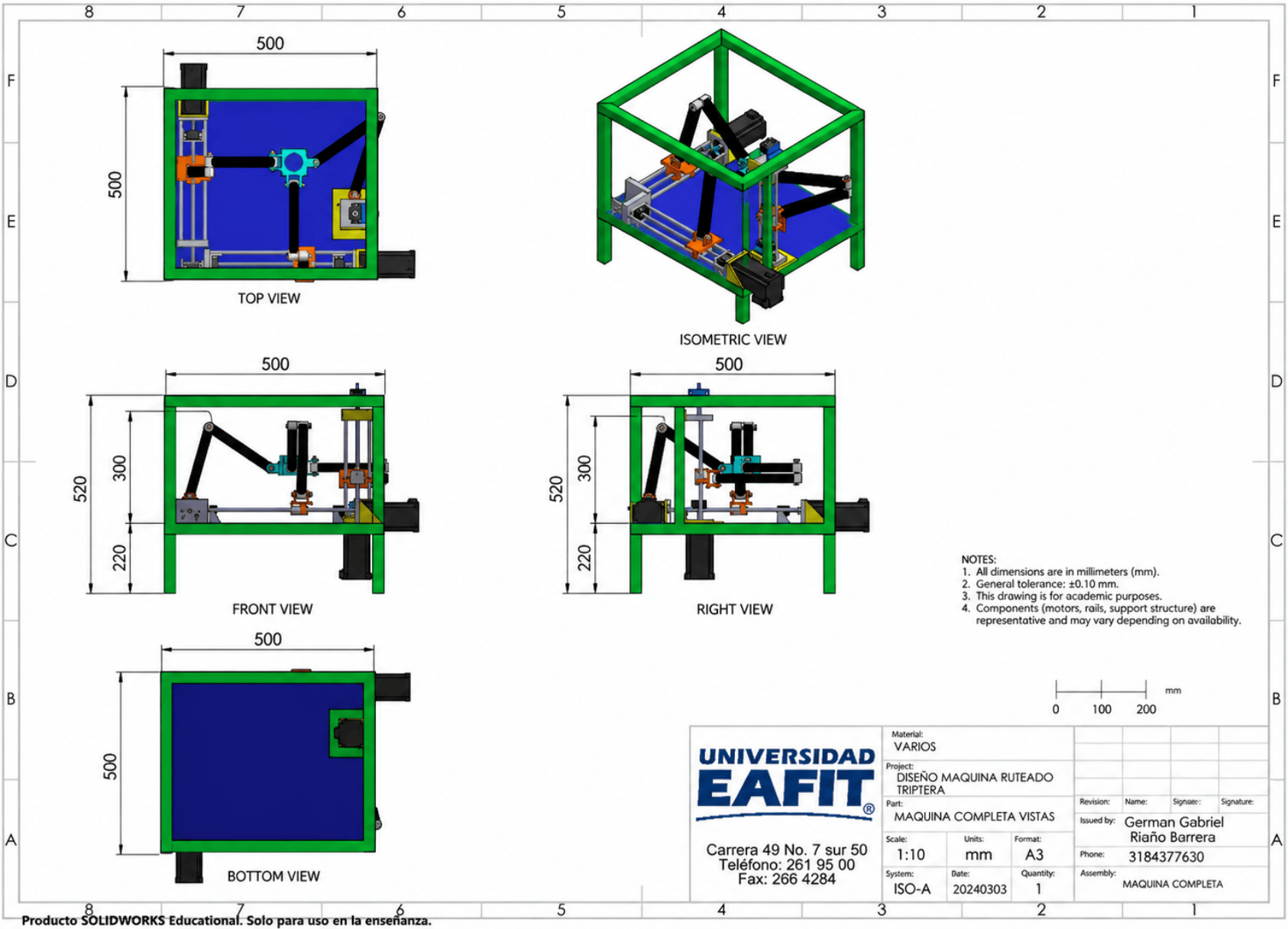


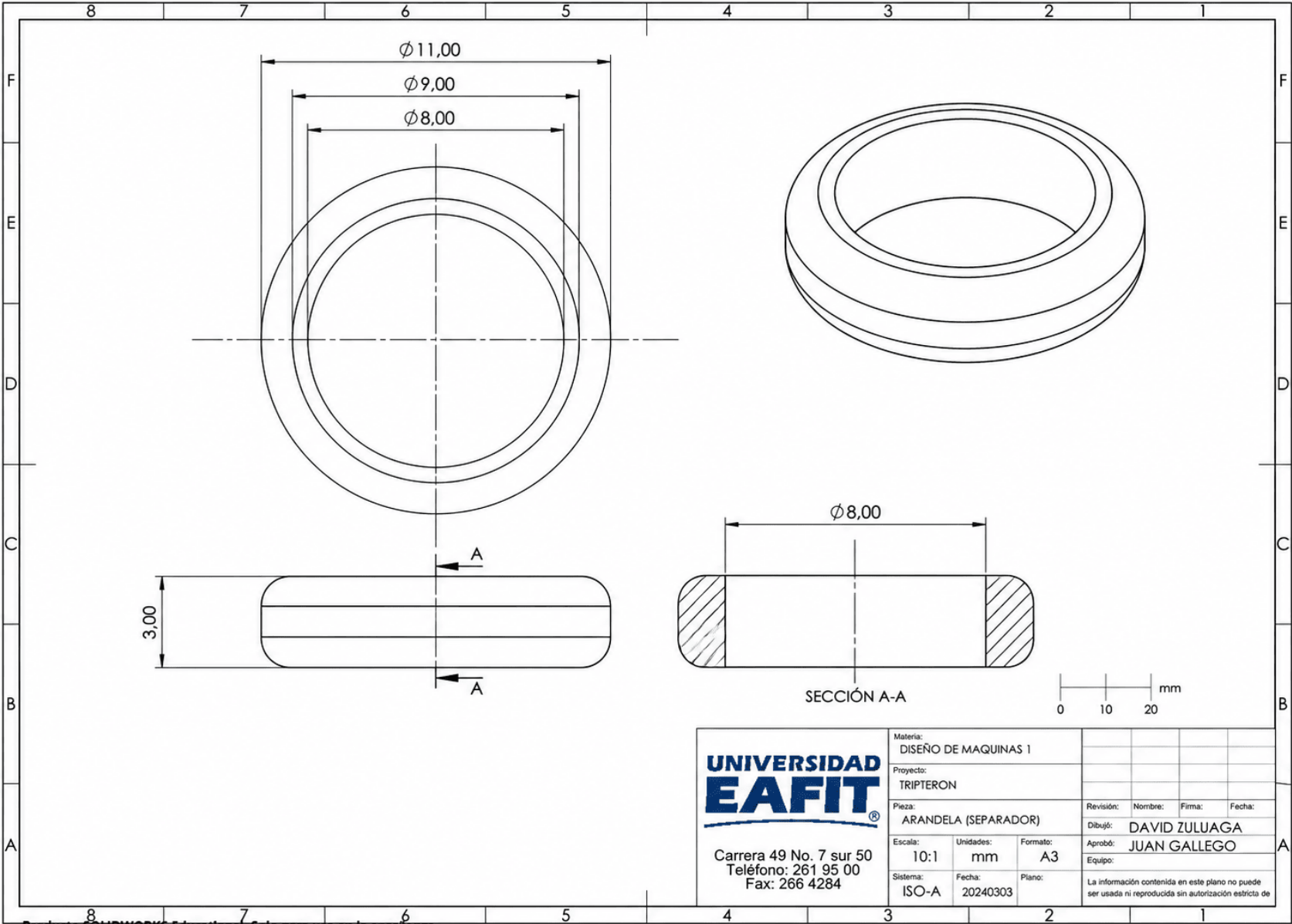






Group 5 – Assembled machine





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Fax: 266 4284

Materia: DISEÑO DE MAQUINAS 1						
Proyecto: TRIPTERON						
Pieza: ARANDELA (SEPARADOR)			Revisión:	Nombre:	Firma:	Fecha:
Escala: 10:1			Dibujó: DAVID ZULUAGA			
Unidades: mm			Aprobó: JUAN GALLEG0			
Formato: A3			Equipo:			
Sistema: ISO-A			La información contenida en este plano no puede ser usada ni reproducida sin autorización estricta de			
Fecha: 20240303						
Plano:						

